

Flow Performance Enhancement of a Double-Feedback Fluidic Oscillator Using Triangular Rib Parameterization

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Abstract

The Coanda surface plays a critical role in governing the oscillation characteristics and flow performance of double-feedback fluidic oscillators. This study systematically investigates the influence of triangular rib parameterization on oscillator performance. Four triangular ribs are introduced on each Coanda surface, and two geometric design strategies are examined. These include a fixed-height (FH) family with half apex angles of 15°, 30°, 45°, and 60°, and a fixed-base (FB) family with half apex angles of 30°, 45°, and 60°. Two-dimensional, unsteady, compressible computational fluid dynamics simulations are performed using air as the working fluid. The Shear Stress Transport (SST) $k-\omega$ turbulence model is employed, and the results are benchmarked against a smooth Coanda surface oscillator. The FH family achieved the highest oscillation frequencies, ranging to improvements of 45.2-47.1% over the smooth design, while reducing the primary recirculation bubble area by approximately 49.4-50.8%. In contrast, the FB family provided greater flexibility in balancing oscillation frequency, jet deflection angle, and pressure loss. Among all configurations, Tri30B-Ribs exhibited the most favorable overall flow performance, combining a 41.0% increase in oscillation frequency and a jet deflection angle of 31.6°, comparable to the smooth configuration (32.0°). The proposed frequency-deflection-pressure ratio (FDPR) identified Tri30B-Ribs as the optimum configuration, achieving the highest FDPR of 1.396 among ribbed designs. These findings demonstrate that triangular rib parameterization enables effective passive tailoring of fluidic oscillator performance for application-specific requirements.

Keywords: fluidic oscillator; triangular ribs; Coanda surface; jet oscillation frequency; jet deflection angle; sweeping jet; recirculation bubble; apex angle

1. Introduction

The fluidic oscillator is a passive fluidic device that generates a high-frequency self-sustained sweeping jet without employing any moving mechanical components. The fluidic oscillator relies entirely on internal flow instabilities to produce periodic jet switching at the exit nozzle. The absence

of moving parts makes the fluidic oscillator mechanically simple, highly reliable, and straightforward to manufacture and integrate into engineering systems. These features, along with its wide frequency range via geometric design, make the fluidic oscillator suitable for many engineering applications. These include active flow and separation control [1–5], acoustic resonance suppression [6], flow measurement and metering [7], passive scalar mixing enhancement [8], and surface heat transfer augmentation through sweeping jet impingement [9, 10]. The oscillating jet formed by a double feedback fluidic oscillator covers a significantly wider target surface area compared to a conventional steady impinging jet. It promotes enhanced turbulent mixing owing to its high oscillation frequency and broad coverage characteristics, which together make it particularly effective for cooling applications on heated surfaces [11, 12].

The operating mechanism of the double feedback oscillator is fundamentally governed by the Coanda effect within the mixing chamber [13, 14]. When a fluid jet enters the mixing chamber through the supply nozzle, even a small flow disturbance creates an asymmetric pressure distribution on the mixing chamber walls. This pressure difference causes the jet to attach to one of the two curved Coanda surfaces, a phenomenon known as the Coanda effect. Once attached, a portion of the main jet is diverted into the feedback channel, which carries low pressure back to the inlet region of the mixing chamber. The feedback flow enters between the jet entering from the inlet and the attached wall, gradually forcing the primary jet away from that wall until it attaches to the other Coanda surface, completing half an oscillation cycle. This self-sustaining process produces a periodic bistable oscillation at the exit nozzle, generating a sweeping jet that alternates between maximum deflection angles on either side of the oscillator centerline.

The influence of oscillator internal geometry on the flow performance parameters, has been the subject of extensive numerical and experimental investigation. Bobusch et al. [15] conducted numerical investigations on geometric parameters of a double feedback oscillator, studying variations in mixing chamber inlet width, exit nozzle width, feedback channel width, feedback channel length, and feedback channel shape. They established that the feedback channel width exerted the most significant influence on oscillation frequency among all parameters studied. In a parametric study by Baghaei and Bergada [16], the oscillation frequency of the jet was most effectively improved by varying the feedback channel width. The other parameters, such as the width of the mixing chamber and its wall's entry and exit angles, produced comparatively smaller effects. A smaller entrance wall width to the mixing chamber was found to produce reverse flow in the feedback channels due to large pressure wave generation, causing significant fluctuations in fluid mass flow rate. Woszidlo et al. [17] investigated the transient flow field of a double feedback oscillator and observed that increasing

the inlet to the mixing chamber increased the output frequency, while making the feedback channels rounded reduced the formation of separation bubbles in the mixing channel.

Slupski and Kara [18] numerically investigated the effect of changes in feedback channel geometry using two-dimensional unsteady compressible flow simulations. Different combinations of feedback channel height and width were tested across various mass flow rates, and the results demonstrated that increasing the feedback channel height raised the oscillation frequency up to a certain limiting value, while increasing the feedback channel width had the opposite effect. Bergada et al. [19] explored the impact of feedback channel length on the amplitude and frequency of fluidic oscillator operation under compressible flow conditions. An increase in feedback channel length resulted in a decrease in the frequency of exit flow oscillation, and comparatively longer feedback channels tended to diminish the starting oscillation, leaving the main jet oscillating at higher intrinsic frequencies within the mixing chamber.

Research into the effect of oscillator inlet and exit nozzle geometry on jet characteristics has also contributed significantly to the field. Sang et al. [20] conducted a numerical investigation of the influence of nozzle throat width and power nozzle dimensions on the characterization of internal flow features, demonstrating that a substantial reduction in throat width relative to the mixing chamber inlet led to a significant increase in the fluid flip time within the oscillator. Jeong and Kim [21] numerically investigated the effect of the ratio of inlet nozzle to splitter-throat width and found that increasing this ratio raised the exit peak jet velocity ratio by 11.8–16.82%. Alam and Kara [22] investigated the effect of geometric design of the exit nozzle on oscillating jet actuator performance. The findings showed that the changes in geometry at the nozzle exit had no significant impact on oscillation frequency but noticeably altered the downstream velocity and pressure distributions. Kim and Kim [23] studied the effects of a bent oscillator outlet on oscillator characteristics both with and without an external crossflow, demonstrating that outlet bending modified both frequency and jet spreading characteristics depending on the bend angle. A parametric numerical study of a fluidic oscillator having a sweeping-vortex low-frequency mode was conducted by Yang et al. [24], who identified several key geometric ratios governing the oscillation mode transitions between different jet deflection states.

Although several geometric parameters, including the feedback channels, mixing chamber, outlet throat, and feedback flow configuration, have been investigated for their effects on fluidic oscillator performance, recent research consistently demonstrates that the Coanda surface exerts the strongest influence on oscillation frequency and jet dynamics. Tajik et al. [25] performed an experimental sensitivity study involving variations in feedback channel width, feedback channel outlet width, and Coanda surface shape. It was observed that Coanda surface shape was the most sensitive parameter,

affecting oscillation frequency among all parameters studied. Similarly, Hossain et al. [26] found that the Coanda surface had a strong effect on the oscillation frequency in their numerical investigation comparing the effects of changes in Coanda surface geometry and the surface geometry of feedback channels in a double feedback oscillator. Nili-Ahmadabadi et al. [27] introduced a novel vortex-based feedback oscillator design that incorporated mixing chamber and consequently Coanda surface modifications and achieved 50% increase in oscillation frequency and a 5% reduction in pressure ratio compared to the conventional design. Tomac and Sundstrom [28] proposed a feedback-type fluidic oscillator with independently adjustable sweep and inclination angles achieved through control flow injection near the throat. Their experimental and numerical results demonstrated a wide range of sweep and inclination angle control with only a modest decrease in oscillation frequency. Song et al. [29] employed pressure-sensitive paint measurements combined with modal analysis to study the spatio-temporal structures and flow mechanisms of double feedback fluidic oscillators. They quantified the direct influence of Coanda surface curvature on the dominant oscillation modes. Different Coanda surface geometries modified the recirculation zones and feedback flow of the main jet, thereby influencing the mechanism of sweeping jet formation. Zhai and Fan [30] conducted experimental and numerical investigations on a double-feedback fluidic oscillator and demonstrated that variations in oscillator thickness and outlet throat width significantly influence the oscillation frequency, outlet tangential velocity, and jet deflection angle. Martinke et al. [31] developed a modified fluidic oscillator with externally controlled feedback flow, allowing independent adjustment of oscillation frequency and flow conditions. Their experiments showed that feedback flow strongly influenced oscillation patterns, whereas maintaining a constant Strouhal number resulted in similar jet dynamics.

While the investigations summarized above demonstrate the critical role of the Coanda surface geometry, they are exclusively concerned with smooth Coanda surface shaping, modifications of surface curvature, inclination, radius, or profile. An alternative and distinctly different approach involves the introduction of discrete surface roughness elements, specifically ribs, directly onto the Coanda surface to modify the jet–wall interaction and recirculation bubble dynamics. This approach was explored by Hussain et al. [32], demonstrating through a numerical study of rectangular ribs having various aspect ratios installed on the Coanda surface that significantly altered the primary recirculation bubble in the mixing chamber. The ribs deflect the attaching jet away from the Coanda surface at each rib tip, compressing the bubble and reducing the resistance to jet switching. The highest aspect ratio case ($AR = 1.56$) achieved a 131% increase in oscillation frequency and a 22% reduction in pressure drop compared to the smooth configuration, with a frequency-deflection-pressure ratio 43% higher than the smooth case. A subsequent investigation [33] systematically examined the influence of the number of rectangular ribs and established that four ribs per Coanda

surface provided the optimal balance between frequency enhancement (146% over the smooth case), jet deflection angle preservation, and pressure drop reduction. The performance parameter FDPR was found to be 87.7% higher for the four-rib configuration compared to the smooth oscillator. Building on these findings, a comprehensive multi-objective optimization study [34] employing integrated CFD–Taguchi–Grey Relational Analysis (GRA) with Analytical Hierarchy Process (AHP) weighting established that the number of ribs contributed 65.01% to oscillation frequency response, rib angle contributed 76.86% to jet deflection angle, and aspect ratio contributed 65.93% to pressure drop reduction. The optimum design attained improvements of 19.9% in oscillation frequency, 39.7% in jet spreading angle, and 17.6% reduction in pressure loss.

Although previous studies have extensively investigated rectangular rib geometries on the Coanda surface, the influence of other rib shapes like triangular-rib geometry and parameterization on the flow performance of fluidic oscillators remains unexplored. In particular, the effects of rib height and rib-face inclination have not been systematically separated to determine their respective roles in modifying jet attachment, recirculation-bubble development, oscillation frequency, jet deflection, and pressure loss. This distinction is important because changing the apex angle of a triangular rib can simultaneously alter its height, base length, and face inclination, making it difficult to attribute observed performance changes to a single geometric feature. Building on the demonstrated effectiveness of discrete Coanda-surface ribs in previous investigations [32–34], the present study introduces triangular ribs and employs two complementary geometric parameterization strategies. In the fixed-height (FH) family, the rib height is maintained constant while the base length and half apex angle are varied, thereby isolating the influence of rib-face inclination at a fixed protrusion height. In the fixed-base (FB) family, the rib base is maintained constant while the rib height and half apex angle are varied, enabling the effect of changing rib protrusion height to be examined. Four ribs are retained on each Coanda surface based on the favorable rib count identified in the authors' previous investigation [33].

The objective of the present study is therefore to establish how these two triangular-rib parameterizations modify the internal unsteady flow structure and the resulting flow performance of the oscillator. Particular attention is given to the evolution of the primary recirculation bubble and its relationship with oscillation frequency, while jet deflection and pressure loss are evaluated to identify the associated performance trade-offs. To provide a unified engineering assessment, frequency-pressure ratio (FPR), jet-deflection-pressure ratio (JDPR), and the combined frequency-deflection-pressure ratio (FDPR) are employed alongside the individual performance parameters. The resulting analysis provides a basis for identifying triangular-rib geometries according to whether the primary design objective is frequency enhancement or balanced oscillation, jet coverage, and pressure

performance. The study addresses three specific questions: (i) whether maintaining rib height or rib base provides greater control over the oscillation dynamics, (ii) how changes in triangular-rib geometry modify the primary recirculation bubble and its relationship with jet-switching frequency, and (iii) whether the frequency gains obtained through Coanda-surface modification can be achieved without compromising jet deflection and pressure efficiency. These questions are examined through a combined analysis of instantaneous velocity, pressure, and vorticity fields and the resulting integral performance parameters.

This paper includes: Section 2 covering the problem formulation and case definitions; Section 3 covers the numerical methodology including governing equations, turbulence model, boundary conditions, mesh independence, and performance parameters; Section 4 covers the results and discussion; and finally Section 5 presents the conclusions.

2 Problem Formulation

2.1 Flow Domain Configuration and Oscillator Geometry

The computational domain for the present study is consistent with that used in prior investigations of the same double feedback sweeping jet oscillator geometry [32-35] and is shown in Figure 1. The domain comprises the full interior of the fluidic oscillator, including the mixing chamber, feedback channels, and exit nozzle, as well as an external flow domain downstream of the oscillator exit through which the sweeping jet propagates. The external flow domain extends sufficiently far in all directions from the oscillator exit to avoid boundary effects on the internal oscillation dynamics. The inlet nozzle is located at the supply end of the mixing chamber, and the pressure inlet boundary condition is applied there. Pressure outlet boundary conditions are applied at the far-field boundaries of the external domain. All oscillator walls are treated as no-slip, adiabatic walls. The oscillator geometry consists of a mixing chamber, an exit nozzle, and two feedback channels. The throat diameter of the exit nozzle (D) is taken as the reference length scale throughout the study. In the present study, the exit throat diameter $D = 6.35$ mm; meanwhile, the feedback channel width and maximum mixing chamber width are 7.43 mm and 21.52 mm, respectively. The mixing chamber inlet width is 10.47 mm and its length is 25.79 mm. These dimensions are maintained unchanged across all configurations so that the effect of Coanda surface rib geometry can be isolated.

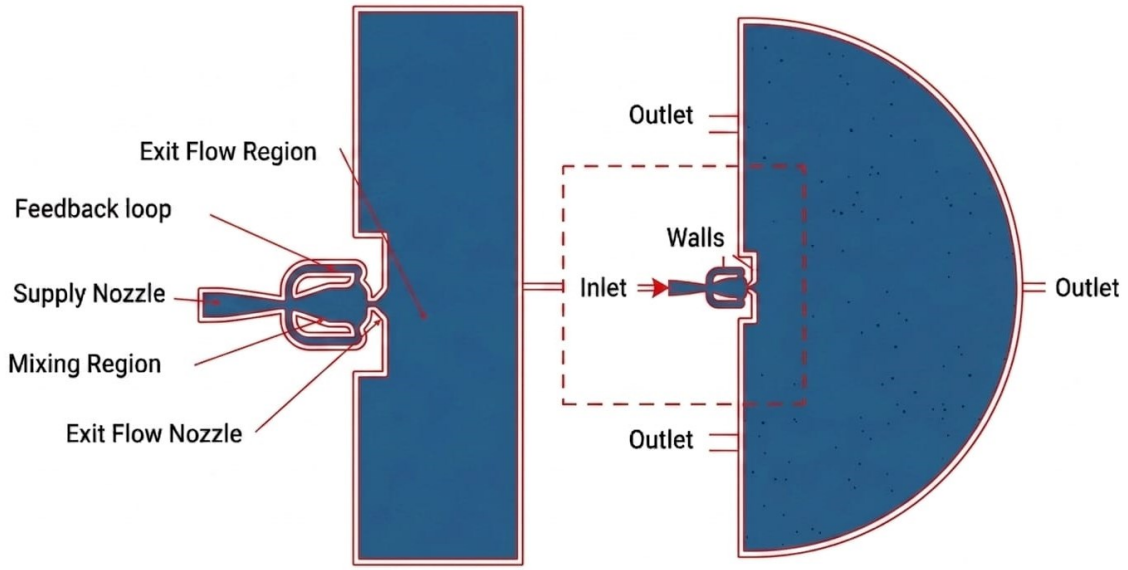


Figure 1. Fluidic oscillator geometric design and computational domain

2.2 Triangular Rib Geometry and Case Definitions

In the present study, the Coanda surface is altered by installing triangular ribs as shown in Figure 2. The cross-sectional geometry of each triangular rib is defined by two parameters: the rib height H , measured perpendicularly from the base of the rib to its apex, and the rib base B , measured along the Coanda surface in the streamwise direction and the half apex angle (α) of the triangular rib is determined by the geometric relationship between these two dimensions as shown Figure 2(b). The orientation of each triangular rib is such that the apex points away from the Coanda surface, directing the incoming flow away from the wall. The pitch between consecutive ribs is set consistently for each configuration.

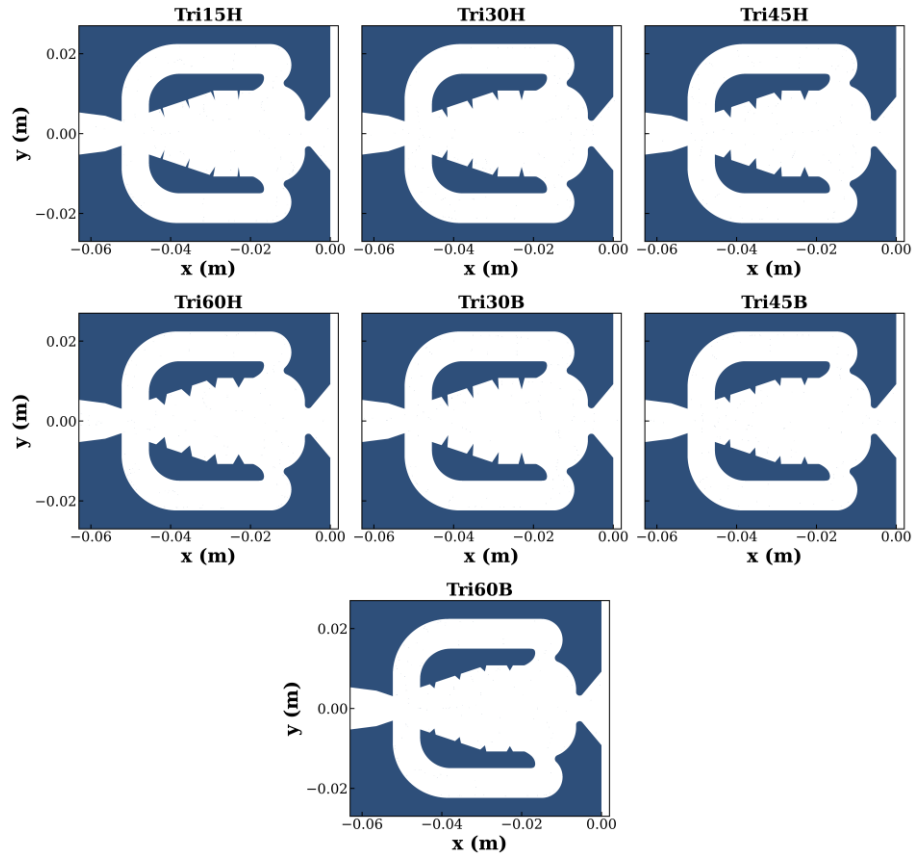
Based on the optimal rib count of four ribs per Coanda surface established in prior rectangular rib investigations [33], all ribbed configurations in the present study employ four triangular ribs per Coanda surface. Two systematic parameterization strategies are employed to investigate the influence of triangular rib apex angle on oscillator performance as shown in Figure 2(a). In the FB family, the rib base B is held constant across all variants, while the rib height H is varied to generate $\alpha = 30^\circ$, 45° , and 60° . By fixing the base length, this strategy isolates the effect of rib height and therefore the degree of jet deflection away from the Coanda surface, allowing the influence of rib protrusion height on recirculation bubble dynamics and oscillation frequency to be examined independently of changes in base length. In the FH family, the rib height H is held constant across all four variants, while the rib base B is varied to generate $\alpha = 15^\circ$, 30° , 45° , and 60° . This strategy isolates the effect of apex angle and thus the inclination of the rib face, at a constant protrusion height, enabling the influence of face angle on jet deflection and Coanda surface interaction to be characterized independently of

changes in rib height. At a fixed base length, achieving a $\alpha = 15^\circ$ requires a substantially taller rib than the 30° , 45° , or 60° cases. In the FB family, this height becomes large enough that the first rib, located nearest the mixing chamber inlet, significantly blocks the incoming jet. Rather than allowing the jet to deflect and attach to the Coanda surface, the oversized first rib obstructs the flow path before attachment can occur. This blockage also disrupts attachment along the subsequent ribs further downstream. Since the sweeping-jet behavior of this oscillator depends entirely on stable jet attachment to the Coanda surface, this blockage suppresses oscillation altogether rather than enhancing it. For this reason, the $\alpha = 15^\circ$ configuration is excluded from the FB family.

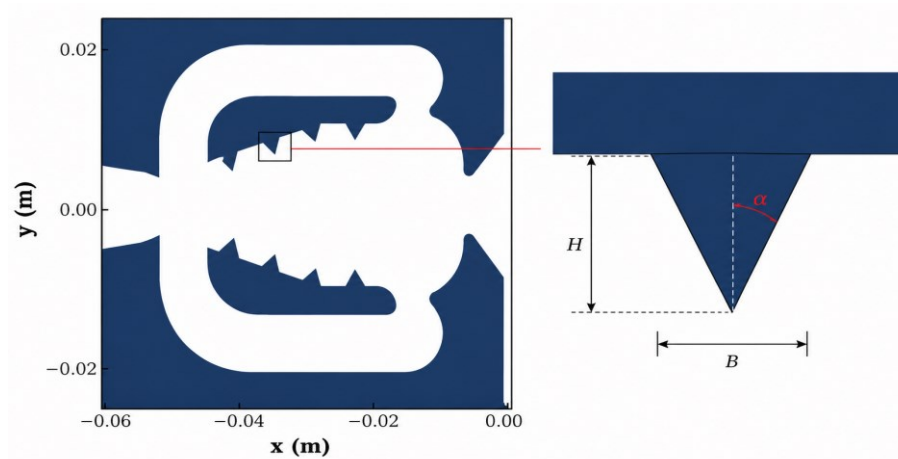
The smooth oscillator (without ribs) serves as the performance benchmark against which all ribbed configurations are evaluated. Table summarizes the complete set of configurations investigated, including rib apex angle, number of ribs per Coanda surface, and the corresponding geometric parameterization strategy. Figure 2 illustrates the geometric construction of the triangular ribs used in the present study. Figure 2(a) presents the complete set of triangular ribbed oscillator geometries for both the FH and FB families, allowing the variation in rib profile across the studied apex angles to be visually compared within and between the two families. Figure 2(b) provides a close-up view of a representative rib, identifying the base (B), height (H), and half apex angle (α) used to define the rib geometry throughout this work.

Table I: Summary of all configurations investigated in the present study

Case ID	Parameterization Strategy	Rib Type	Half Apex Angle (α)	No. of Ribs	Rib Height (H)	Rib Base (B)
Smooth	Baseline	No ribs	—	0	—	—
Tri15H-Ribs	Fixed height	Triangular (FH)	15	4	Fixed	Variable
Tri30H-Ribs	Fixed height	Triangular (FH)	30	4	Fixed	Variable
Tri45H-Ribs	Fixed height	Triangular (FH)	45	4	Fixed	Variable
Tri60H-Ribs	Fixed height	Triangular (FH)	60	4	Fixed	Variable
Tri30B-Ribs	Fixed base	Triangular (FB)	30	4	Variable	Fixed
Tri45B-Ribs	Fixed base	Triangular (FB)	45	4	Variable	Fixed



(a)



(b)

Figure 2: Triangular rib geometry details: (a) Fixed-Height family and Fixed-Base family, (b) close-up view with H and B dimensions

3 Numerical Methodology

3.1 Governing Equations

According to the results of Kruger et al. [36], a two-dimensional investigation of the fluidic oscillator is adequate to resolve the essential qualitative features and quantitative characteristics of the internal flow field and its oscillatory dynamics. Consequently, air is used as the working fluid, and two-dimensional unsteady Reynolds-Averaged Navier-Stokes (URANS) equations are employed to simulate the compressible turbulent flow of air in the fluidic oscillator's interior and its external vicinity. The internal and external fluid flows are governed by the compressible forms of the continuity, momentum, and energy transport equations. To accurately model variable-density turbulent flow, Favre averaging is performed for velocity, temperature, enthalpy, and internal energy terms, while Reynolds averaging is performed for density and pressure. For a detailed discussion of the Favre-averaged formulation, the reader is referred to Oz and Kara [35].

3.1.1 Continuity Equation

$$\frac{\partial(\bar{\rho})}{\partial t} + \frac{\partial(\bar{\rho}\tilde{u}_i)}{\partial x_i} = 0 \quad (1)$$

In Eq. (1), $\tilde{u}_i$ represents the density-weighted (Favre-averaged) velocity components, x_i represents the spatial coordinates, and $\bar{\rho}$ denotes the mean density, which is calculated as a function of time using the ideal gas law.

3.1.2 Momentum Equation

$$\frac{\partial(\bar{\rho}\tilde{u}_i)}{\partial t} + \frac{\partial(\bar{\rho}\tilde{u}_i\tilde{u}_j)}{\partial x_j} = -\frac{\partial\bar{P}}{\partial x_i} + \frac{\partial[2\bar{\nu}_T\bar{S}_{ij} - \overline{\rho u_i''u_j''}]}{\partial x_j} \quad (2)$$

Here $\bar{P}$ represents the mean pressure, $\bar{\nu}_T$ represents the temperature-dependent kinematic viscosity, and $\bar{\nu}_T = \mu_T/\bar{\rho}$, while μ_T is calculated using Sutherland's law [37] as follows:

$$\mu_T = \mu_0 \left(\frac{T}{T_0}\right)^{3/2} \left(\frac{T_0 + S_1}{T + S_1}\right) \quad (3)$$

where the reference viscosity $\mu_0 = 1.72 \times 10^{-5}$ kg/m·s, the fluid temperature T , the reference temperature $T_0 = 273.16$ K, and the effective temperature $S_1 = 110.56$ K. The term $\bar{S}_{ij}$ represents the mean strain-rate tensor and $\overline{\rho u_i''u_j''}$ is the Reynolds stress tensor.

3.1.3 Energy Equation

$$\begin{aligned} \frac{\partial}{\partial t} \left[\bar{\rho} \left(\tilde{e} + \frac{\tilde{u}_i \tilde{u}_i}{2} \right) + \left(\frac{\overline{\rho u_i'' u_i''}}{2} \right) \right] + \frac{\partial}{\partial x_j} \left[\bar{\rho} \tilde{u}_j \left(\tilde{h} + \frac{\tilde{u}_i \tilde{u}_i}{2} \right) + \tilde{u}_j \frac{\overline{\rho u_i'' u_i''}}{2} \right] = \frac{\partial}{\partial x_j} \left[-k \frac{\partial \tilde{T}}{\partial x_j} - \overline{\rho u_j'' h''} + \overline{t_{ji} u_i''} - \right. \\ \left. \overline{\rho u_j'' \frac{1}{2} u_i'' u_i''} \right] + \frac{\partial}{\partial x_j} \left[\tilde{u}_i \left(2 \bar{v}_T \bar{S}_{ij} - \overline{\rho u_i'' u_j''} \right) \right] \end{aligned} \quad (4)$$

where $\tilde{e}$ represents the specific internal energy and $\tilde{h}$ represents the specific enthalpy.

3.2 Turbulence Model

The Reynolds stresses are modeled using the eddy viscosity approach. For this purpose, the Shear Stress Transport (SST) k - ω turbulence model, originally developed by Menter [38] and further validated for industrial applications by Menter et al. [39], is employed, where k is the turbulent kinetic energy and ω is the specific dissipation rate. The SST k - ω model combines the near-wall formulation of the k - ω model with the k - ε behavior in the outer flow region through blending functions, providing improved performance for flows involving adverse pressure gradients and flow separation. These characteristics make the model particularly suitable for the present fluidic oscillator, where strong shear layers, adverse pressure gradients, flow separation, and the formation and evolution of the primary recirculation bubble occur along the Coanda surfaces and within the mixing chamber. The computational mesh was refined to maintain $y^+ < 0.2$ throughout the computational domain, allowing the near-wall region to be directly resolved without the use of wall functions.

For the SST k - ω model implemented in ANSYS FLUENT [40], the turbulent viscosity is calculated as:

$$\mu_t = \frac{\rho k}{\omega} \frac{1}{\max \left(\frac{1}{\alpha^*}, \frac{F_2 S}{a_1 \omega} \right)} \quad (5)$$

where F_2 is the second blending function, S is the strain-rate magnitude, and $a_1 (= 0.31)$ is a closure constant of the SST model. The coefficient α^* is a low-Reynolds-number damping function applied to the turbulent viscosity within the general form of the SST model implemented in ANSYS FLUENT [40]; it is expressed as a function of the local turbulent Reynolds number, $Re_t = \frac{\rho k}{(\mu \omega)}$, approaching unity in the fully turbulent core flow and falling below unity within the viscosity-affected near-wall layer, where it damps the eddy viscosity smoothly toward the wall. The Low-Re Corrections option was not activated for the present computations, so that α^* retained its default value of unity throughout the solution domain, and Equation (5) is thereby consistent with the high-Reynolds-number eddy-

viscosity relation validated for industrial flows by Menter et al. [39]. The complete form of Equation (5) is retained here for fidelity to the governing equations solved by the CFD code.

3.3 Numerical Schemes and Boundary Conditions

Two-dimensional unsteady compressible turbulent flow simulations are performed using the ANSYS FLUENT (version 2022) software package, with air as the working fluid. A second-order accurate upwind scheme for the convective part of the momentum, turbulent kinetic energy, and specific dissipation rate transport equations is employed to minimize numerical diffusion. The pressure and velocity coupling is solved using the Semi-Implicit Method for Pressure Linked Equations (SIMPLE) algorithm. A fully implicit second-order temporal scheme is utilized for time marching to ensure temporal accuracy throughout the unsteady simulation. The simulations are initialized through a set of boundary conditions at the inlet. These conditions specify that the static pressure and temperature are set at 101,325 Pa and 298.16 K, respectively. The simulation is set up with a pressure inlet boundary condition, with $Re = 114,600$ based on the exit nozzle throat diameter D and exit throat velocity V_e ($Re = V_e D / \nu$). The computational setup was first run for 10^4 time-steps (totaling 0.05 s), each with a time-step size of $\Delta t = 5 \mu s$, to achieve a fully developed oscillating jet with stable periodic behavior. After obtaining the stabilization phase, data sampling for statistical analysis was performed by further simulating 10^4 additional time-steps using the same time-step size of $5 \mu s$, covering a 0.05-second sampling interval, to examine and quantify the bistable oscillation characteristics.

3.4 Temporal Convergence Analysis

A temporal sensitivity analysis was carried out to verify that the selected temporal resolution accurately captures the unsteady oscillatory behavior of the fluidic oscillator. Simulations were performed using four different time-step sizes, as summarized in Table . The results indicate that decreasing the time-step size from $5.0 \mu s$ to $2.5 \mu s$ produces a negligible change of only 0.003% in the predicted oscillation frequency, demonstrating that the numerical solution is temporally converged. Consequently, a time-step size of $5.0 \mu s$ was adopted for all subsequent transient simulations.

Table II: Temporal convergence analysis for different time-steps

Time-step Size (μs)	Time Interval for Data Saving (ms)	Frequency (Hz)	Percentage Change in Frequency
10.0	50	425.42	0.099%
7.5	50	425.28	0.066%
5.0	50	425.00	—
2.5	50	424.99	(0.002%)

3.5 Mesh Independence and Model Validation

3.5.1 Mesh Independence

A mesh independence analysis was performed to ensure that the numerical results are free from spurious effects due to domain discretization. For this purpose, four different computational meshes, Coarse (N20), Normal (N40), Fine (N60), and Finer (N80), were created using the ANSYS FLUENT (version 2022) meshing tool, where the subscript denotes the number of elements across the exit nozzle throat diameter D . The boundary layer near the fluidic oscillator walls is resolved with 20 inflation layers normal to the wall, with a growth rate of 1.1 and a first layer thickness of 0.0127 mm (equal to $D/500$). Three concentric spheres of influence with radii of $8D$, $20D$, and $100D$ were used to control element size grading in the external flow domain, with corresponding element sizes of $D/5$ and $D/2.5$ in the outer regions as shown in Figure 3. For the innermost region, the element sizes for the N20, N40, N60, and N80 meshes were $D/20$, $D/40$, $D/60$, and $D/80$, respectively.

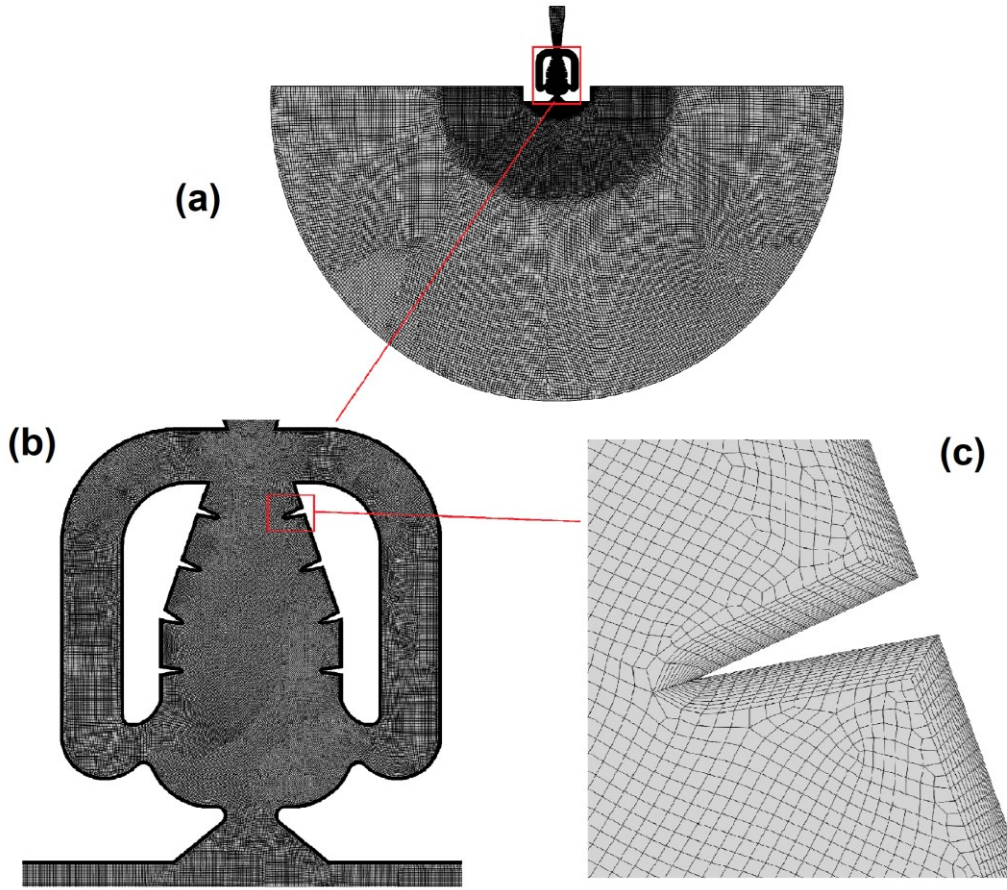


Figure 3: Computational mesh employed for the numerical simulations: (a) overall computational domain, (b) structured hexahedral mesh in the fluidic oscillator region, and (c) close-up view of the high-quality boundary-layer mesh around a triangular rib.

The oscillation frequency of the fluidic oscillator was determined from the centerline velocity signal sampled 6 mm downstream of the nozzle exit. The results of the mesh independence study are summarized in Table . The predicted oscillation frequency increases from 412 Hz for the coarse N20 mesh to 425 Hz for the N40 mesh, indicating that the coarse discretization is unable to fully resolve the flow structures governing the self-sustained oscillation. Further mesh refinement to N60 and N80 produces frequencies of 427 Hz for both meshes, with a maximum variation of only 0.47% relative to the N40 solution, demonstrating that the numerical solution has essentially reached mesh independence. Since the improvement obtained with the finer meshes is negligible while requiring a substantially higher computational cost, the N40 mesh was selected for all subsequent simulations as the optimal balance between numerical accuracy and computational efficiency.

Table III: Mesh independence study results

Mesh	Elements across D	Frequency (Hz)	Difference (%)	Selected
N20 (Coarse)	20	412	—	No
N40 (Normal)	40	425	3.16	Yes
N60 (Fine)	60	427	0.47	No
N80 (Finer)	80	427	0.00	No

3.5.2 Model Validation

The accuracy of the numerical methodology is validated by comparing the oscillation frequency obtained from the present two-dimensional (2D) Computational Fluid Dynamics (CFD) approach against available experimental and numerical results from the literature for the same oscillator geometry. It can be observed from Table that the oscillation frequency obtained by the present 2D numerical methodology (355 Hz) shows good agreement with the experimental results of Slupski et al. [41] (346 Hz), the 2D numerical results of Jurewicz and Kara [42] (347 Hz) and Alam and Kara [22] (350 Hz), and the three-dimensional (3D) numerical results of Oz and Kara [35] (341 Hz). The relative errors are 2.6%, 0.3%, 1.2%, and -1.4% respectively. These small errors confirm the reliability and accuracy of the present 2D numerical methodology in capturing the dominant oscillatory dynamics of the double feedback fluidic oscillator. It should be noted that while 2D numerical models perform comparably well with 3D models in predicting oscillation frequency, a 3D simulation would be required for a more detailed characterization of secondary vortical structures and spanwise flow effects within the mixing chamber.

Table IV: Validation of numerical methodology

Reference	Analysis Type	Frequency (Hz)	Error (%)
Slupski et al. [41]	Experimental	346	—
Jurewicz and Kara [42]	2D Numerical	347	0.3
Alam and Kara [22]	2D Numerical	350	1.2
Oz and Kara [35]	3D Numerical	341	-1.4
Present study	2D Numerical	355	2.6

3.6 Performance Evaluation Parameters

To comprehensively assess the flow performance of each fluidic oscillator configuration, the following performance parameters are defined and evaluated. The oscillation frequency was determined from the time history of the velocity magnitude sampled at the centerline of the oscillator,

6 mm downstream of the exit nozzle. The frequency was evaluated from the period of the complete and repeatable jet oscillation rather than from spectral peaks obtained using Fast Fourier Transform (FFT). The time interval corresponding to one complete oscillation cycle was obtained from the alternating extrema of the velocity signal. This time interval was taken as the oscillation period (T), and the oscillation frequency was calculated as ($f = 1/T$). Thus, the reported frequency represents the complete cyclic motion of the jet. The jet deflection angle (θ) is calculated from the jet half-width δ , defined as the lateral distance y from the centerline at which the time-averaged velocity magnitude equals half the maximum centerline velocity, measured at a downstream distance $x = 3D$ from the oscillator exit [25]:

$$\theta = \tan^{-1} \left(\frac{\delta}{x} \right) \quad (6)$$

The frequency-pressure ratio (FPR) relates the oscillation frequency gain to the corresponding pressure drop across the oscillator [32]:

$$FPR = \frac{f_e/f_s}{\left(\Delta p_e/\Delta p_s \right)^{1/3}} \quad (7)$$

The jet deflection angle–pressure ratio (JDPR) relates the jet deflection angle change to the pressure drop [32]:

$$JDPR = \frac{\theta_e/\theta_s}{\left(\Delta p_e/\Delta p_s \right)^{1/3}} \quad (8)$$

The combined frequency-deflection-pressure ratio (FDPR) is the primary combined performance parameter, simultaneously accounting for oscillation frequency, jet deflection coverage, and pressure loss [32]:

$$FDPR = \frac{f_e/f_s * \theta_e/\theta_s}{\left(\Delta p_e/\Delta p_s \right)^{1/3}} \quad (9)$$

where the subscripts e and s denote the enhanced (ribbed) and smooth (no ribs) cases respectively, f_e and f_s are the oscillation frequencies, θ_e and θ_s are the jet deflection angles, and Δp_e and Δp_s are the

pressure drops across the oscillator. A higher FDPR value indicates a better combined flow performance relative to the smooth baseline. The Strouhal number (St) is defined as:

$$St = \frac{f * D}{V_{exit}} \quad (10)$$

where V_{exit} is the time-averaged velocity at the exit nozzle throat.

4 Results and Discussion

The double feedback fluidic oscillator produces a self-sustained sweeping jet through the bistable attachment and periodic switching of the primary jet within the mixing chamber. To assess the flow performance of each configuration investigated in the present study, the parameters: oscillation frequency, jet deflection angle, and pressure drop are examined. The oscillation frequency governs the temporal uniformity and intensity of jet sweeping, while the deflection angle determines the lateral coverage of the sweeping jet in the downstream field. The results for all oscillators having triangular ribs on the Coanda surface are investigated and compared against the smooth oscillator baseline.

4.1 Effects of Triangular Rib Geometry on Internal Flow Dynamics

When the jet enters the mixing chamber, even a minor flow disturbance creates an asymmetric pressure distribution along the walls. This pressure difference triggers the Coanda effect, which makes the jet end up attaching to one of the two curved surfaces. As the jet advances toward the wall, the Coanda effect intensifies, and attachment becomes complete, corresponding to the phase angle $\Phi = 0^\circ$. A fraction of the jet simultaneously enters the feedback channel, where it propagates back towards the main jet entering the mixing chamber from the inlet. This jet progressively pushes the jet until it attaches to the opposite wall at $\Phi = 180^\circ$, completing half an oscillation cycle. The phase angles $\Phi = 0^\circ, 45^\circ, 90^\circ, 135^\circ$, and 180° capture this half-period evolution, after which the process repeats symmetrically to produce a self-sustained oscillating jet at the exit nozzle. The primary recirculation bubble that forms in the mixing chamber on the side opposite to jet attachment is the key flow feature governing the switching speed and therefore the oscillation frequency. The size and evolution of this bubble across the five phase angles for all cases in the present study is examined through the streamline, pressure contour, and vorticity magnitude plots presented in Figure 4 through Figure 9.

Figure 4 presents the velocity contour plots for the smooth case and the three fixed-base configurations at all five phase angles. For the smooth oscillator at $\Phi = 0^\circ$, a large primary recirculation bubble occupies a substantial portion of the mixing chamber on the side opposite to jet

attachment. The smooth Coanda surface allows the primary jet to follow the wall curvature closely, generating prolonged, strong wall adhesion. The feedback flow must overcome this adhesion before switching can occur. The large recirculation bubble and the wall adhesion result in a strong resistance to the jet switching and hence the lowest oscillation frequency among all cases. For the Tri30B-Ribs configuration, the triangular ribs at $\alpha = 30^\circ$ cause the jet to periodically detach from the wall at each rib apex. At $\Phi = 0^\circ$, the recirculation bubble in Tri30B-Ribs is visibly reduced compared to the smooth case. At a fixed base length, the $\alpha = 30^\circ$ half-apex angle corresponds to the tallest rib height among the three fixed-base configurations, and this pronounced rib height deflects the primary jet away from the Coanda surface strongly enough to disrupt sustained wall attachment and compress the recirculation bubble. As the α increases to 45° and 60° in Tri45B-Ribs and Tri60B-Ribs, the rib height reduces progressively at the fixed base length, leaving more open space within the mixing chamber flow passage. With shorter ribs, the deflecting effect on the primary jet is weaker. The periodic detachment forced at each rib apex is therefore less pronounced. The jet stays attached to the Coanda surface for a longer time. Consequently, the recirculation bubble re-establishes itself over a larger portion of the cycle than in the Tri30B-Ribs case. As rib height decreases, this disruption effect progressively weakens. This explains why the oscillation frequency drops in the FB family for α above 30° . The behavior gradually trends back toward the slower, smooth-case-like response.

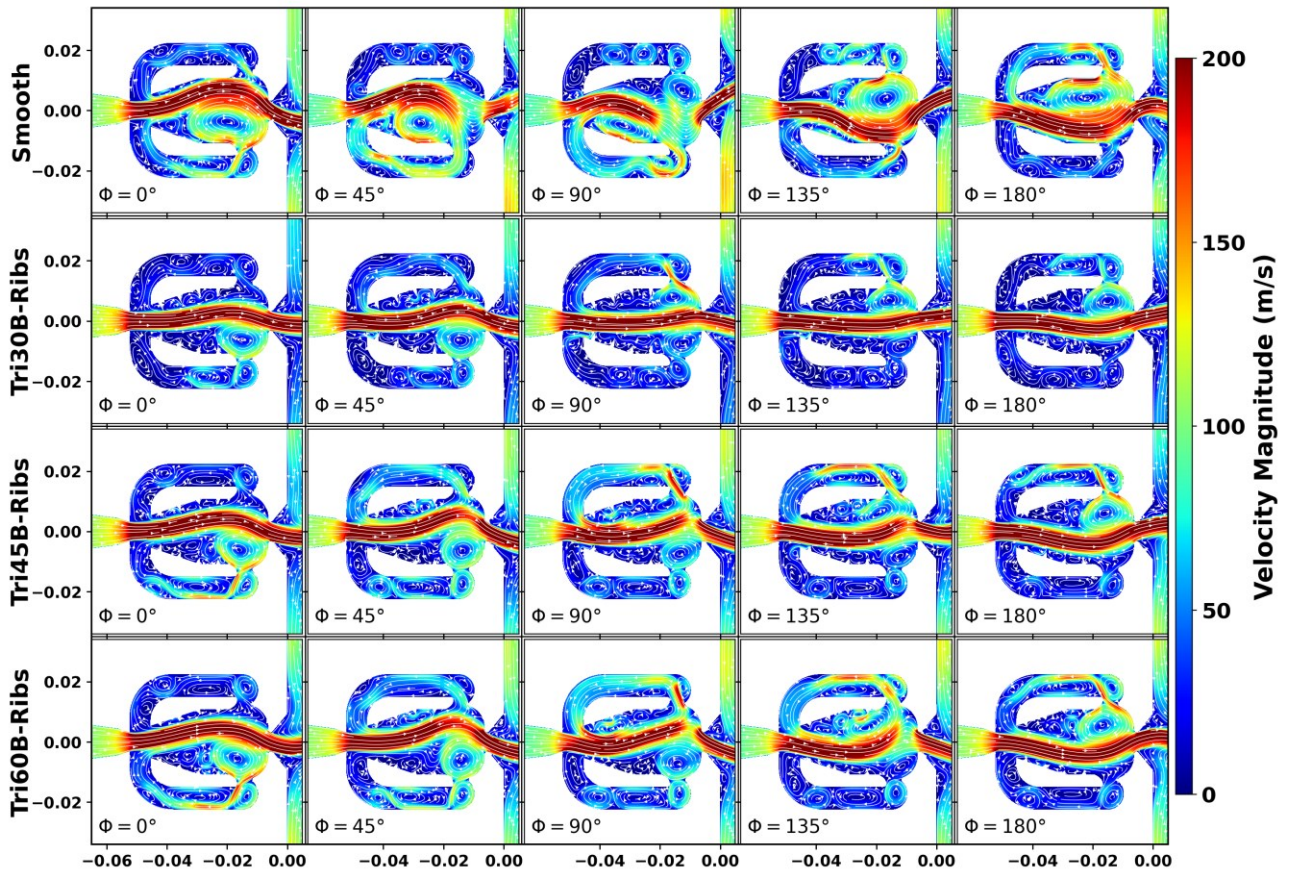


Figure 4: Velocity contour plots (velocity magnitude, m/s) for smooth case and fixed-base (FB) configurations at phase angles $\Phi = 0^\circ$, 45° , 90° , 135° , and 180°

Figure 5 presents the velocity contour plots for the smooth case and the four fixed-height configurations. Since the rib height H is held constant across all FH variants, the degree to which each rib deflects the primary jet away from the Coanda surface is essentially the same for all four cases. The recirculation bubble at $\Phi = 0^\circ$ is uniformly compressed across $\alpha = 15^\circ$ - 60° , representing Tri15H-Ribs, Tri30H-Ribs, Tri45H-Ribs, and Tri60H-Ribs cases. The bubble becomes consistently and substantially smaller than in the smooth case. The principal geometric difference between FH variants is that at $\alpha = 15^\circ$, the rib base is narrower and the face is more steeply inclined. It produces a sharper but more localized jet redirection. As α increases toward 60° , the base widens, and the face becomes more gradually inclined, producing a gentler but more distributed jet redirection. Despite this geometric difference, the streamline plots reveal that the recirculation bubble compression is nearly identical across all four FH configurations at every phase angle. This confirms that rib height, not face inclination, is the dominant parameter governing recirculation bubble dynamics when height is fixed.



Figure 5: Velocity contour plots (velocity magnitude, m/s) for smooth case and fixed-height (FH) configurations at phase angles $\Phi = 0^\circ, 45^\circ, 90^\circ, 135^\circ$, and 180°

The pressure contour plots in Figure 6 for the smooth and FB cases reinforce the streamline observations. For the smooth oscillator, a large prominent low-pressure zone (deep blue region) is visible in the mixing chamber at $\Phi = 0^\circ$, representing the primary recirculation bubble. For Tri30B-Ribs, this low-pressure zone is almost diminished, and the pressure distribution across the mixing chamber is more uniform. As the α increases to 45° and 60° (Tri45B-Ribs and Tri60B-Ribs), the mixing chamber develops alternating high- and low-pressure patches near the ribs. Hence, a larger low-pressure region re-emerges inside the mixing chamber toward the feedback channel entrance in the Tri60B-Ribs case. This re-emergence is consistent with the shorter ribs channeling excess jet momentum into the feedback loop. It weakens the main jet and allows the recirculation bubble to persist and expand, reducing the oscillation frequency. At later phase angles, the pressure evolution across all FB cases follows the same qualitative pattern as the smooth case, with reduced magnitude of the low-pressure zone for ribbed cases.

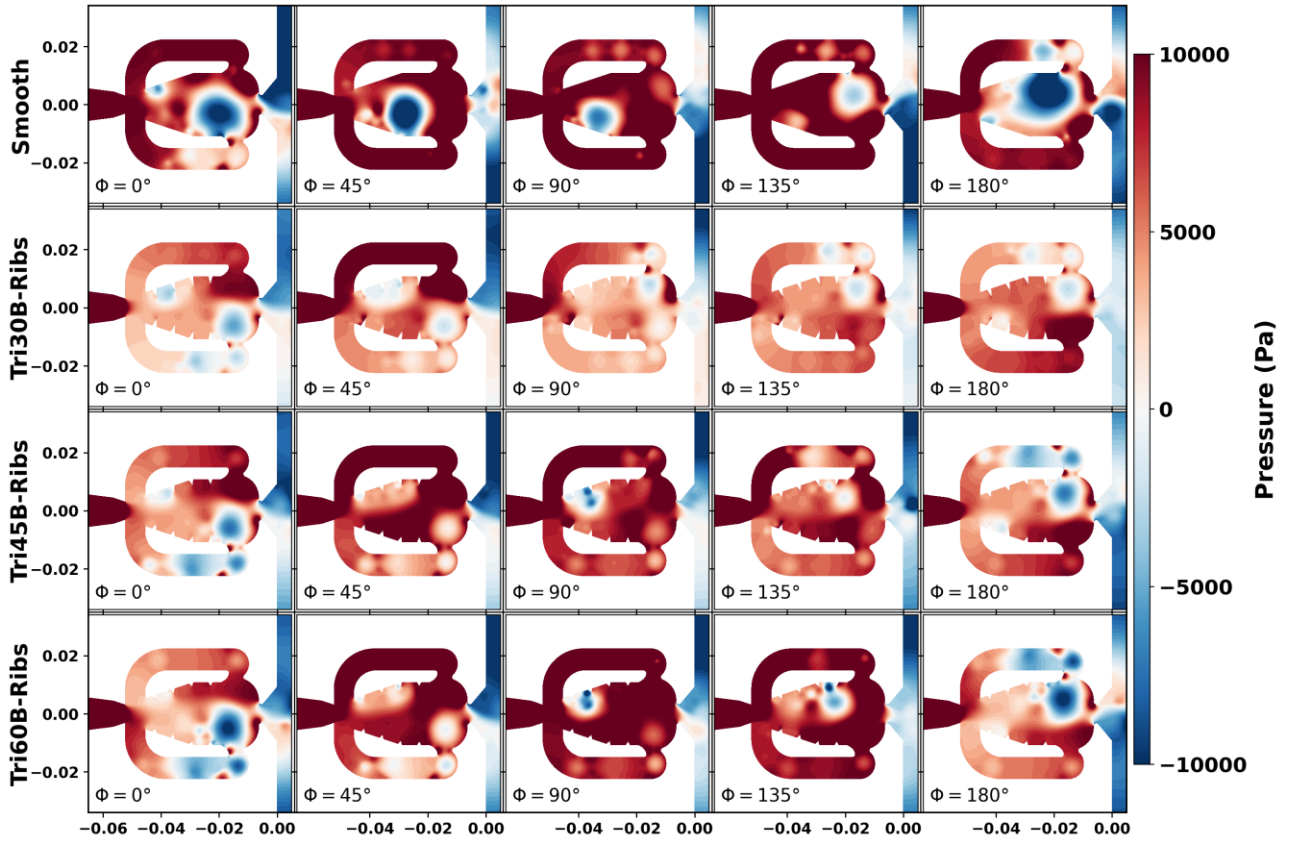


Figure 6: Pressure contours (Pa) for smooth case and fixed-base (FB) configurations at phase angles $\Phi = 0^\circ, 45^\circ, 90^\circ, 135^\circ$, and 180°

Figure 7 presents the instantaneous pressure contours for the smooth and FH cases at five representative oscillation phases. Compared to the smooth oscillator, all four FH configurations exhibit significantly reduced low-pressure zones in the mixing chamber at $\Phi = 0^\circ$. This indicates that the triangular ribs suppress the extent of the large separation zone formed over the Coanda surfaces. The pressure distributions across all four FH variants remain remarkably similar at every phase angle. This further confirms that, for a fixed rib height, the apex angle has only a minor influence on the overall pressure field. Slight local variations can be observed in the immediate vicinity of the rib surfaces. These differences do not alter the global pressure distribution or the oscillation characteristics. In all FH cases, the mixing chamber exhibits a more uniform pressure field, and the low-pressure separation region remains consistently smaller than that of the smooth case throughout the oscillation cycle.

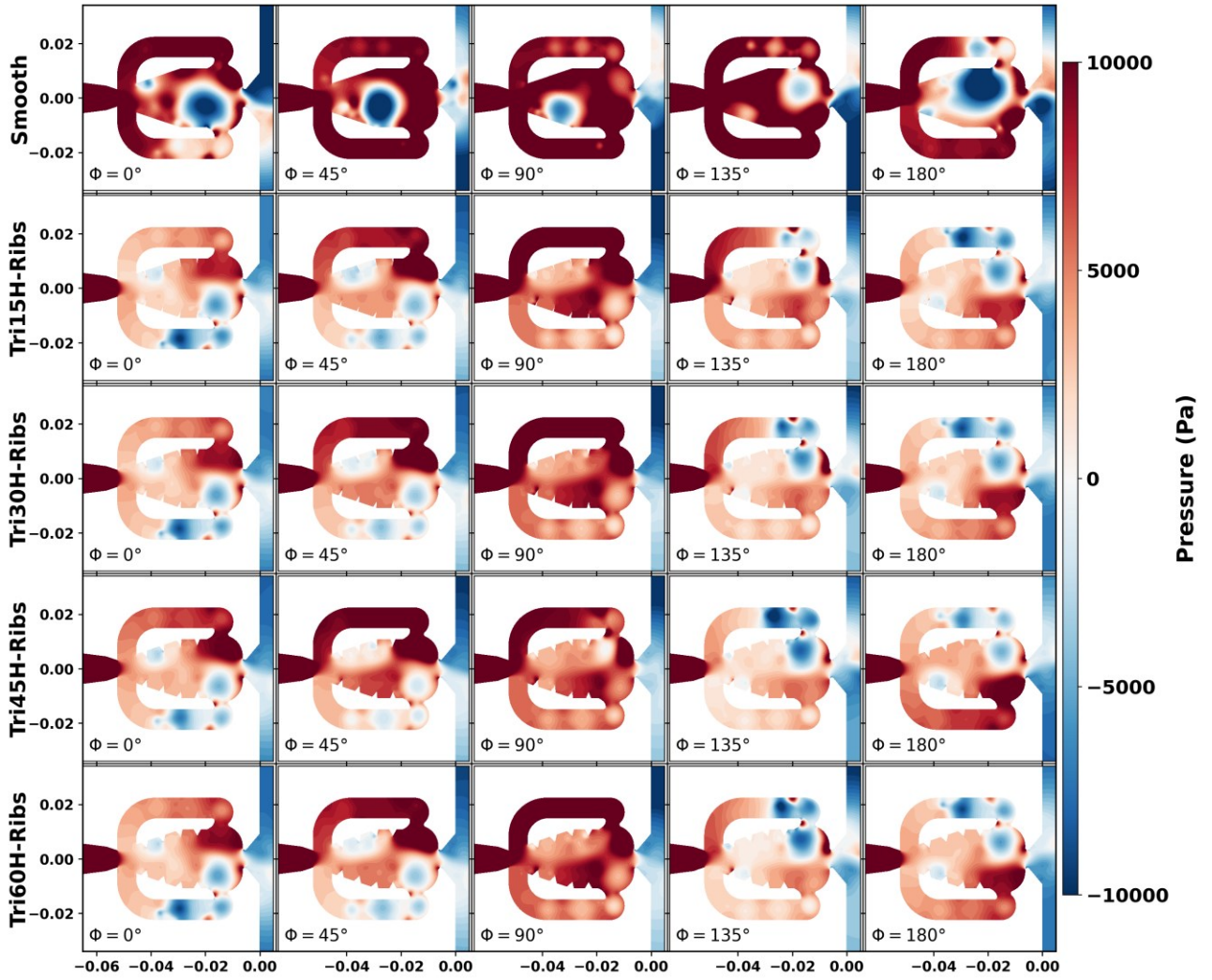


Figure 7: Pressure contours (Pa) for smooth case and fixed-height (FH) configurations at phase angles $\Phi = 0^\circ, 45^\circ, 90^\circ, 135^\circ$, and 180°

The vorticity magnitude contours for the smooth and FB cases in Figure 8 further highlight the influence of the triangular ribs on the flow structure throughout the oscillators. For the smooth oscillator at $\Phi = 0^\circ$, the two high-vorticity shear layers generated by the primary jet are concentrated on one side of the mixing chamber. On the other side, a relatively large low-vorticity region occupies the recirculation zone within the mixing chamber. In contrast, the ribbed configurations exhibit a more spatially distributed vorticity field. They have localized high-vorticity regions developing around the rib surfaces and along the Coanda walls. Among the ribbed configurations, $\alpha = 30^\circ$ (Tri30B-Ribs) exhibits the most fragmented vorticity field, with numerous localized high-vorticity structures distributed throughout the mixing chamber. The smallest low-vorticity recirculation region, reflecting the stronger flow disturbance produced by the relatively taller ribs. As the α increases to 45° and 60° (Tri45B-Ribs and Tri60B-Ribs), the rib height decreases, reducing the intensity of the rib-induced vortical structures. This allows the low-vorticity recirculation region to progressively enlarge. Consequently, the vorticity field becomes increasingly smoother, with the Tri60B-Ribs case

exhibiting a flow pattern that trends toward the smooth oscillator while still retaining the beneficial mixing effects introduced by the triangular ribs.

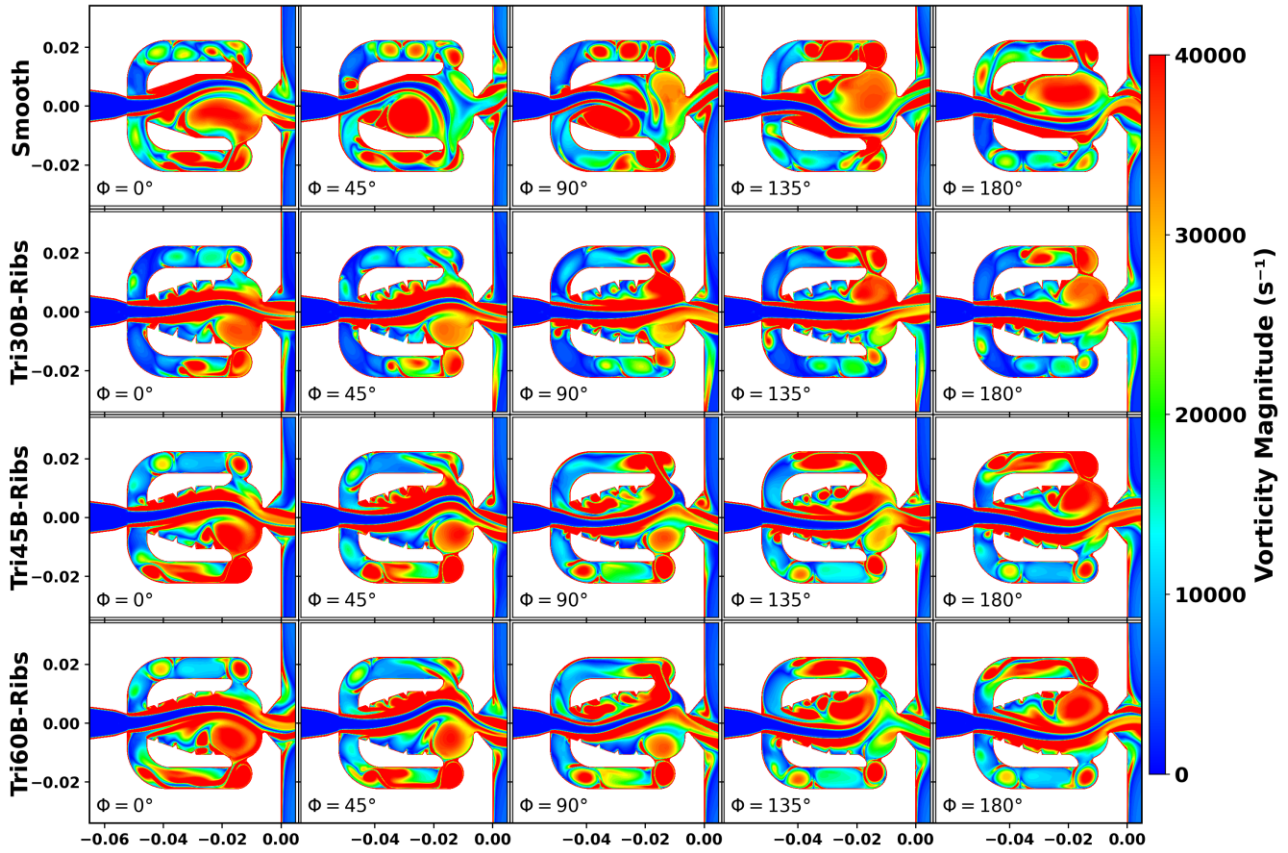


Figure 8: Vorticity magnitude contours (s^{-1}) for smooth case and fixed-base (FB) configurations at phase angles $\Phi = 0^\circ, 45^\circ, 90^\circ, 135^\circ$, and 180°

Figure 9 presents the vorticity contours for the smooth and FH cases. Compared with the smooth oscillator, all four FH configurations exhibit a substantially modified vorticity field within the mixing chamber. In the smooth case, the primary jet is bounded by two continuous high-vorticity shear layers. They enclose a relatively large low-vorticity recirculation region. In contrast, the introduction of triangular ribs produces numerous localized high-vorticity regions along the ribs and Coanda walls. It results in a more uniformly distributed vorticity field and a noticeable reduction in the size of the central low-vorticity region. The four FH configurations exhibit remarkably similar vorticity distributions throughout the oscillation cycle, indicating that varying the apex angle at a fixed rib height has only a minor influence on the overall flow structure. Only subtle local differences are observed in the vicinity of the rib surfaces. The large-scale vorticity patterns remain nearly unchanged. Furthermore, the vorticity contours at $\Phi = 90^\circ$ - 180° show that the high-vorticity shear layers occupy a larger portion of the mixing chamber than in the smooth case. This is consistent with the enhanced mixing characteristics and the faster jet switching observed for all FH configurations.

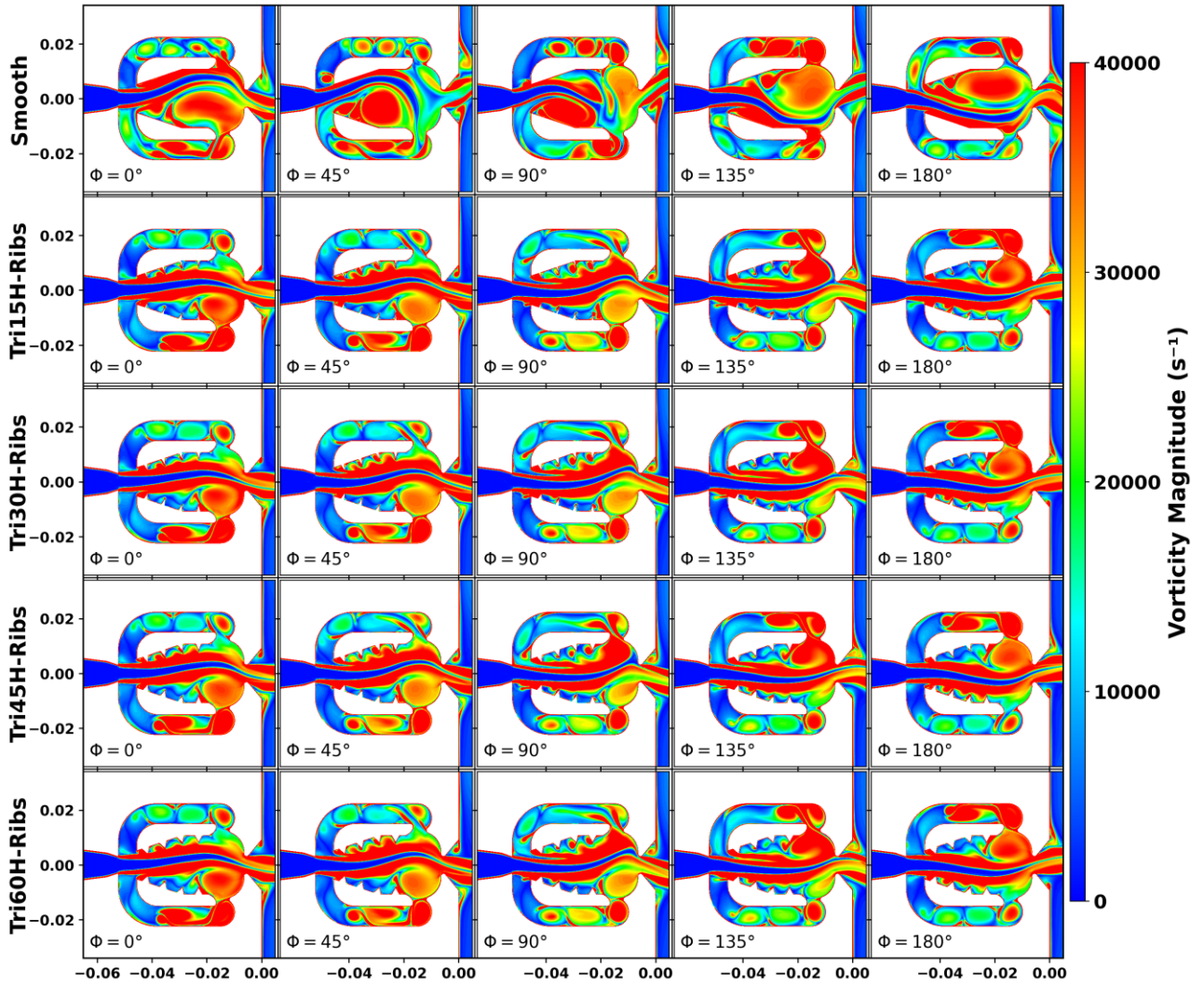


Figure 9: Vorticity magnitude contours (s^{-1}) for smooth case and fixed-height (FH) configurations at phase angles $\Phi = 0^\circ, 45^\circ, 90^\circ, 135^\circ$, and 180°

4.2 Oscillation Frequency and Frequency-Pressure Ratio

Figure 10 presents the oscillation frequency results for all seven ribbed configurations, with the smooth baseline (425 Hz) shown as a dashed reference line. The trends observed in Figure 10 directly reflect the recirculation bubble dynamics discussed in Section 4.1. As demonstrated by the velocity plots, oscillation frequency is governed primarily by the ability of the rib geometry to compress the primary recirculation bubble. It weakens the Coanda attachment, thereby shortening the jet switching period.

The FB family exhibits a pronounced dependence of oscillation frequency on apex angle. Tri30B-Ribs achieves the highest frequency within this family, reaching 599 Hz, corresponding to a 41.0% improvement over the smooth case. However, increasing the α to 45° and 60° reduces the oscillation frequency to 497.5 Hz and 474 Hz, representing improvements of only 17.1% and 11.5%, respectively. As discussed in Section 4.1, maintaining a constant rib base while increasing the apex

angle progressively reduces the rib height. Consequently, the ability of the ribs to periodically interrupt wall attachment diminishes, allowing the primary jet to remain attached to the Coanda surface for a longer duration. This permits the primary recirculation bubble to recover progressively toward the smooth-case behavior. The enlarged recirculation bubble delays the pressure feedback required for jet switching, resulting in a longer switching period and hence a lower oscillation frequency. The monotonic decline in frequency therefore provides quantitative confirmation of the velocity contour observations presented in Figure 4. It demonstrates that the reduction in rib height dominates the flow behavior in the fixed-base design.

In contrast, the FH family maintains consistently high oscillation frequencies across the entire range of $\alpha = 15^\circ$ - 60° . Tri15H-Ribs, Tri30H-Ribs, Tri45H-Ribs, and Tri60H-Ribs achieve frequencies of 617, 625, 621, and 625 Hz, respectively. The total variation across all four configurations is only 8 Hz despite a 45° variation in α . This confirms that the oscillation frequency is only weakly affected by face inclination when rib height remains fixed. As shown by the velocity contour plots in Figure 5, all FH configurations produce nearly identical compression of the primary recirculation bubble throughout the oscillation cycle. As α increases, the rib face inclination changes from a sharper, localized deflection to a gentler, more distributed redirection of the jet. Yet these geometric differences do not significantly alter the overall reduction in wall attachment or the size of the recirculation bubble because the rib height remains unchanged. Consequently, the feedback mechanism responsible for jet switching operates with nearly identical effectiveness in all FH configurations, producing oscillation frequencies within a narrow 8 Hz band. These results confirm that, when the rib height is fixed, apex angle primarily modifies the local jet redirection without significantly influencing the global switching dynamics. Rib height therefore emerges as the dominant geometric parameter governing oscillation frequency.

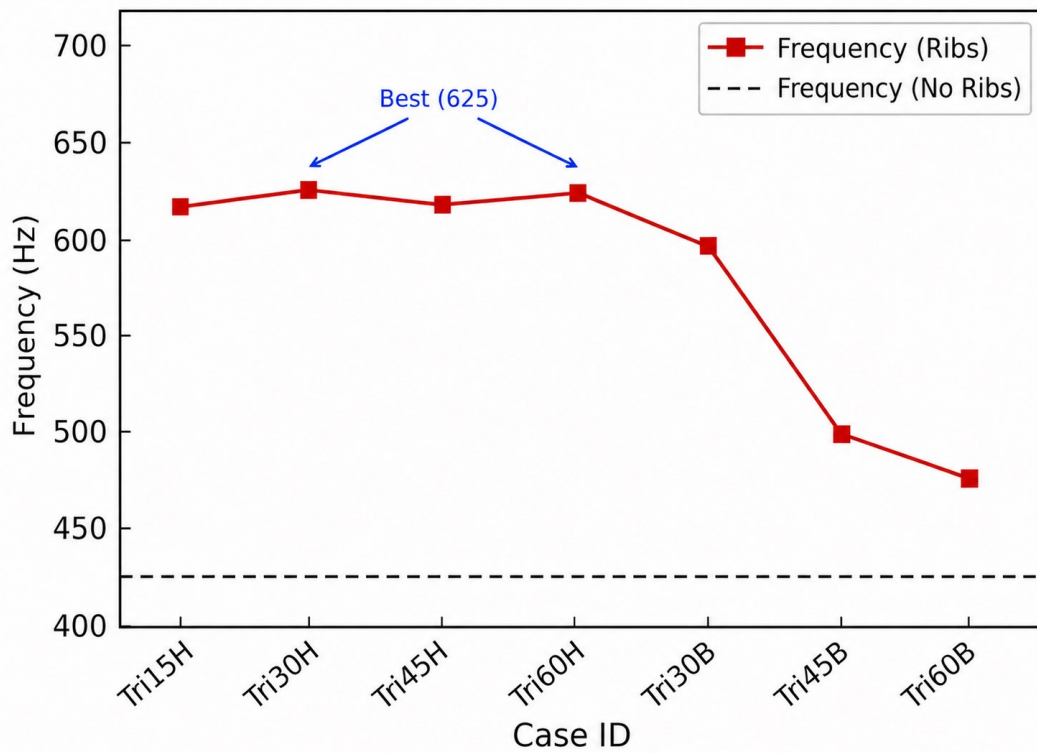


Figure 10: Oscillation frequency for all ribbed configurations; dashed line shows smooth case baseline

Figure 11 presents the frequency-pressure ratio (FPR), which normalizes the frequency improvement by the cube root of the pressure drop ratio relative to the smooth baseline. The FH family achieves FPR values of 1.4531 (Tri15H), 1.4724 (Tri30H), 1.4626 (Tri45H), and 1.4725 (Tri60H), all substantially above the smooth case of 1.0. These values confirm that the FH family achieves its frequency enhancement with only a modest accompanying pressure drop, making it the most frequency-efficient family in terms of pressure loss. Tri30B-Ribs achieves an FPR of 1.4149, still significantly above the smooth baseline. Tri45B-Ribs and Tri60B-Ribs show progressively reduced FPR values of 1.1771 and 1.1249, reflecting their reduced frequency gains despite their lower pressure drops. The FPR analysis confirms that the FH family and Tri30B-Ribs represent the most pressure-efficient configurations for oscillation frequency enhancement.

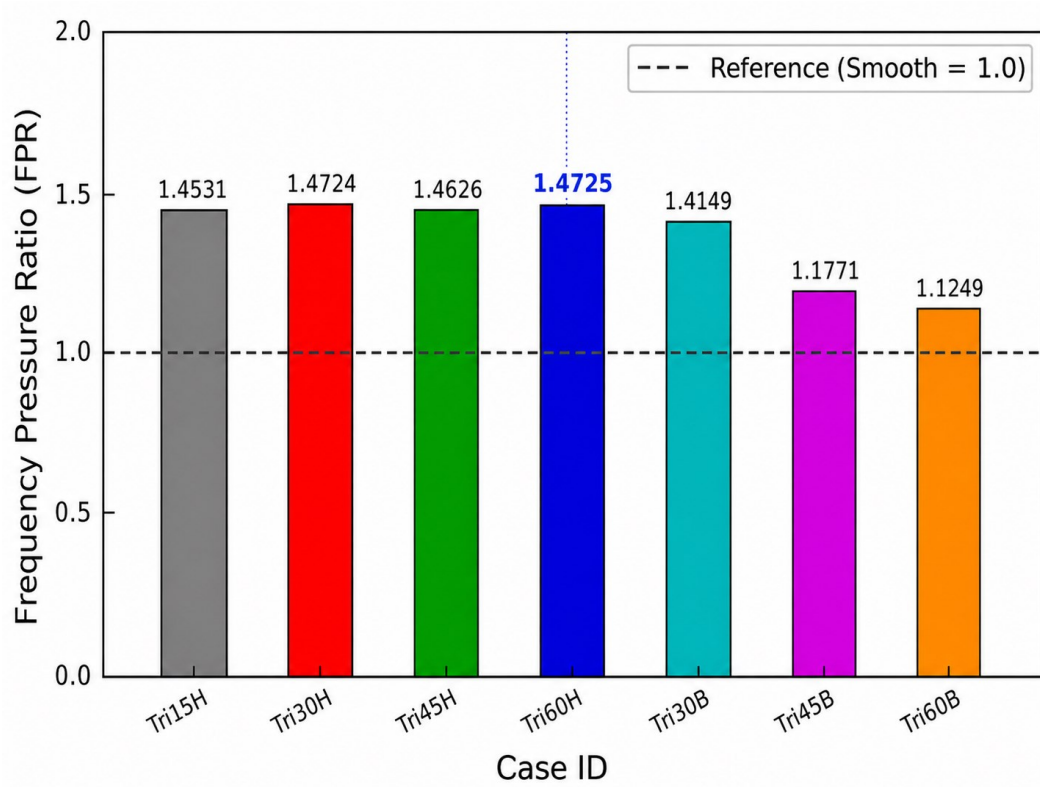


Figure 11: Frequency-Pressure Ratio (FPR) for all ribbed configurations; dashed line shows smooth case reference

4.3 Primary Recirculation Bubble Analysis

The primary recirculation bubble was identified from the velocity contour plots at $\varphi = 0^\circ$ for all eight configurations. The bubble appears as an enclosed low-velocity region within the mixing chamber, as illustrated in Figure 12. Its cross-sectional profile was found to be approximately elliptical in all cases. Accordingly, the streamwise length (L) and wall-normal width (W) were measured at the maximum extents of the bubble using the probe tool in Tecplot 360, as indicated in Figure 12, and the projected area was calculated using the ellipse area formula:

$$A = (\pi/4) \times L \times W \quad (11)$$

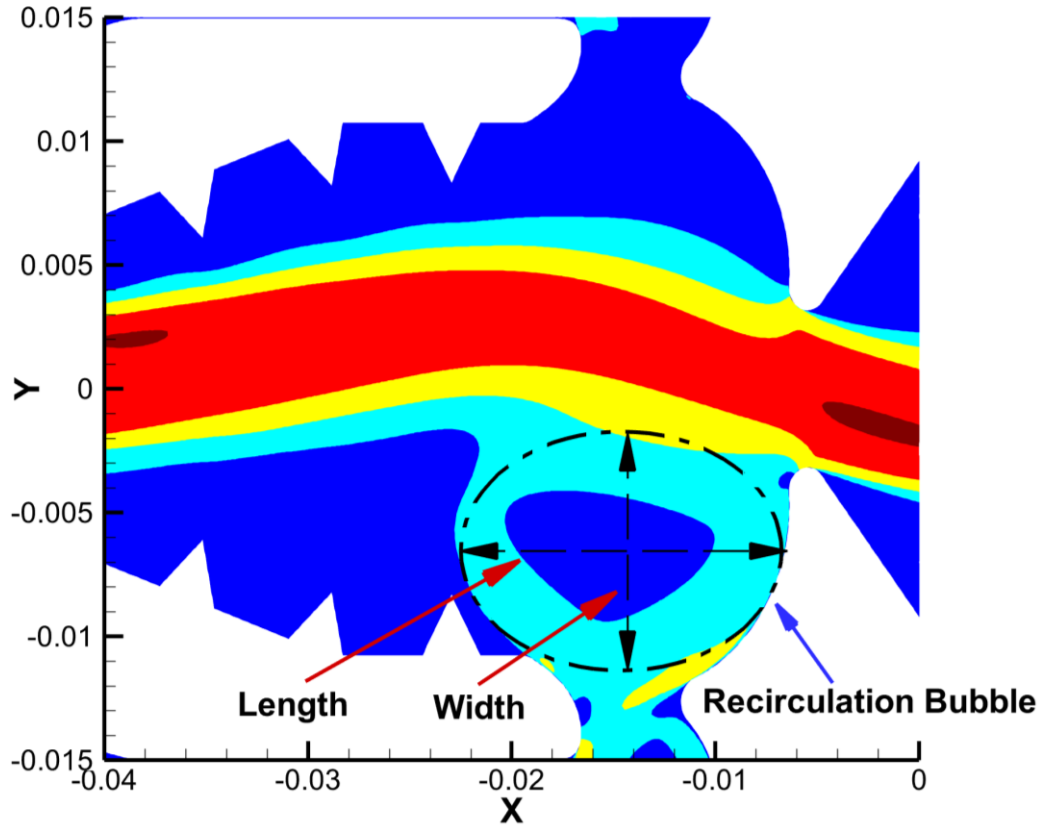


Figure 12: Recirculation Bubble Size on Coanda Surfaces Phase Angle $\phi = 0^\circ$

The variation in recirculation bubble dimensions across all configurations at $\phi = 0^\circ$ is presented in Figure 13. The Smooth configuration produced the largest recirculation bubble, with an area of 231.57 mm². It occupies a substantial portion of the mixing chamber and restricts lateral jet switching between the Coanda surfaces. The introduction of triangular ribs substantially reduced the recirculation bubble area in all ribbed configurations, with reductions ranging from approximately 37.2% to 50.8% relative to the Smooth case. In the fixed-base (B) family, the bubble area decreased from 145.47 mm² for Tri60B to 122.15 mm² for Tri30B as the apex angle decreased. This indicates that smaller apex angles are more effective in reducing the recirculation bubble within this family. The corresponding bubble areas represent reductions of approximately 37.2–47.3% relative to the Smooth case.

The FH family exhibited bubble areas within a narrow range of 113.95–117.14 mm², corresponding to reductions of approximately 49.4–50.8% relative to the Smooth configuration. The small variation among the four fixed-height cases indicates that the apex angle has only a limited influence on the recirculation bubble when the rib height is fixed. In comparison, the fixed-base family exhibited a wider range of bubble areas, reflecting a stronger dependence on apex angle. These results suggest that rib height exerts a stronger influence on the recirculation bubble size than the apex angle. In contrast, variations in apex angle produce only minor changes when the rib height is fixed.

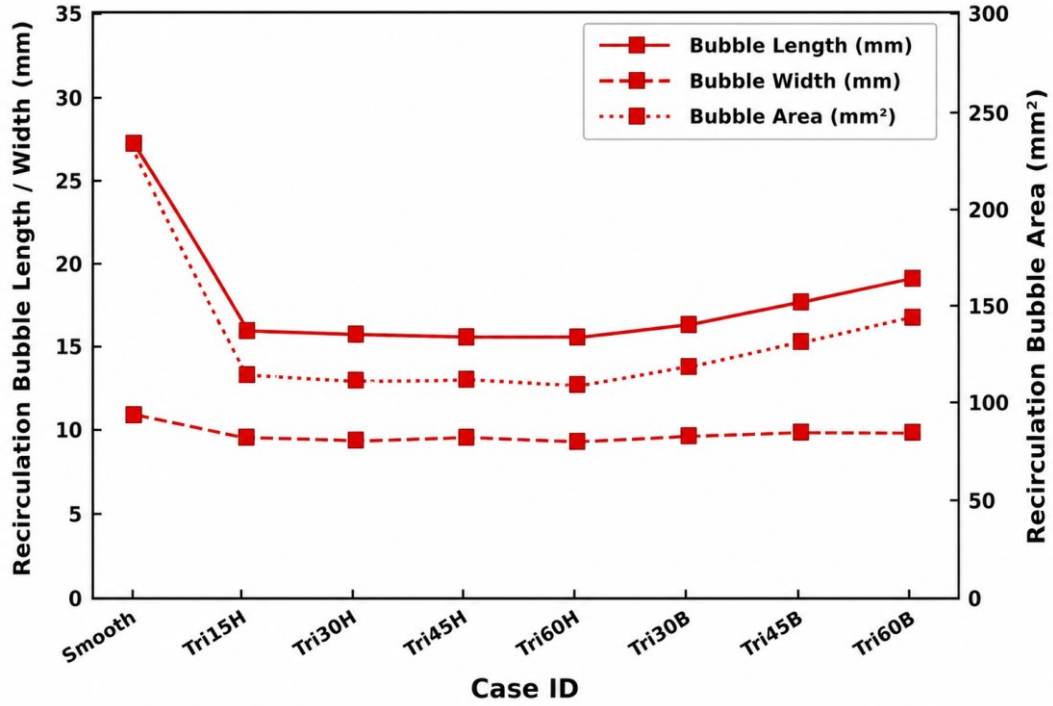


Figure 13: Recirculation Bubble Size across the oscillator for configurations

The relationship between the recirculation bubble area and oscillation frequency is presented in Figure 14. An inverse relationship is observed across all configurations, with smaller recirculation bubble areas corresponding to higher oscillation frequencies, while larger areas correspond to lower frequencies. The Smooth configuration, which exhibits the largest recirculation bubble (231.57 mm²), produced the lowest oscillation frequency of 425 Hz. The introduction of triangular ribs reduced the recirculation bubble area and was accompanied by a corresponding increase in oscillation frequency. This shows that suppression of the primary recirculation bubble facilitates more frequent jet switching within the mixing chamber.

The relationship also reveals two distinct response regimes. The FH family is clustered within a narrow range of recirculation bubble areas (113.95–117.14 mm²) and oscillation frequencies (617–625 Hz). It reveals that variations in apex angle have only a limited influence when the rib height is maintained constant. In contrast, the FB family exhibits a wider spread in both recirculation bubble area (122.15–145.47 mm²) and oscillation frequency (599–474 Hz). As the apex angle increases within this family, the accompanying increase in bubble area is associated with a progressive reduction in oscillation frequency. It demonstrates a greater sensitivity of the flow dynamics to geometric variations when the rib height changes simultaneously.

Although the overall trend is inverse, the relationship is not strictly linear. The fixed-height configurations occupy a plateau region where relatively small changes in recirculation bubble area

produce only marginal differences in oscillation frequency. In contrast, the fixed-base configurations exhibit a steeper decline in frequency with increasing bubble area. This behavior suggests that once the recirculation bubble is sufficiently suppressed, further reductions in its size provide only limited additional enhancement in oscillation frequency. Meanwhile, larger recirculation bubbles increasingly inhibit lateral jet switching and consequently reduce the oscillation frequency.

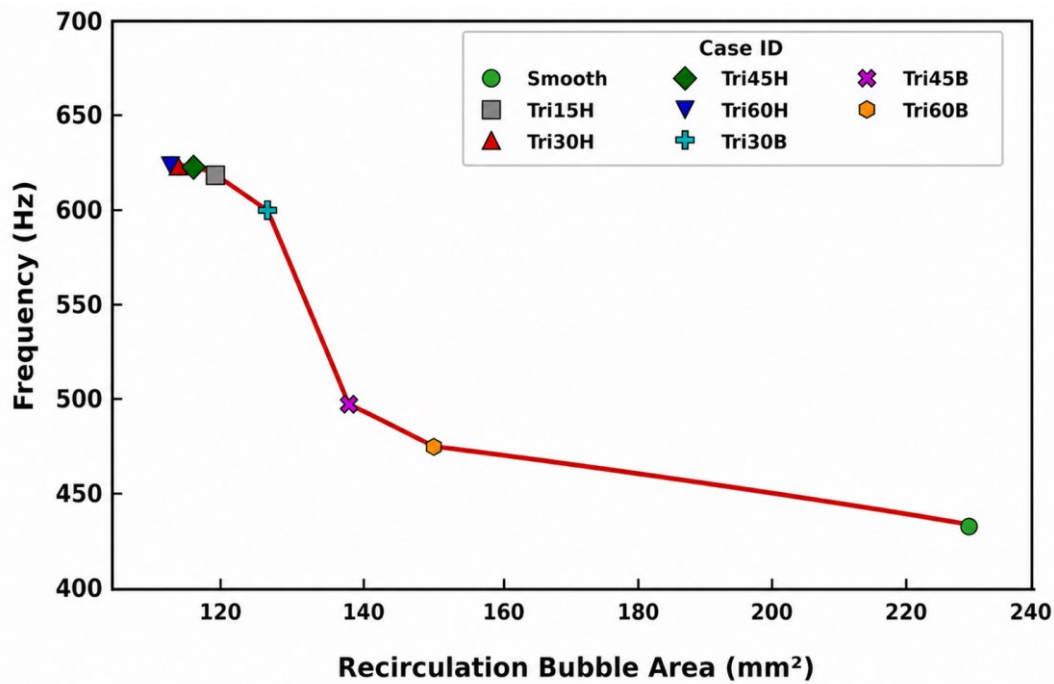


Figure 14: Effect of Recirculation Bubble Area on Oscillation Frequency

4.4 Pressure Drop and Strouhal Number

Figure 15 presents the oscillator pressure drop for all ribbed and the smooth configurations. All seven ribbed configurations achieve a reduction in pressure drop relative to the smooth baseline. Within the FH family, the pressure drop values are tightly clustered in the range of 33,177-33,212 Pa, all slightly below the smooth case value of 33,306 Pa. The low variation within the FH family reflects the dominant influence of the constant rib height: since H is unchanged across all FH variants, the degree of jet-wall contact and the associated viscous friction drag remain essentially constant, producing negligible variation in pressure drop across the family. The fixed-base family exhibits a much more pronounced and monotonically decreasing pressure drop trend. Tri30B-Ribs achieves 32,923 Pa, Tri45B-Ribs achieves 32,753 Pa, and Tri60B-Ribs achieves the minimum pressure drop of the entire study at 32,464 Pa, a 2.53% reduction below the smooth case. This progressive reduction is attributed to the increasingly early jet detachment from the Coanda surface as rib height grows with apex angle

in the FB family, which substantially reduces the direct jet–wall contact area and the associated friction drag. While this pressure reduction is beneficial in isolation, it is accompanied by the frequency penalties discussed in Section 4.2, representing a clear trade-off within the FB parameterization strategy.

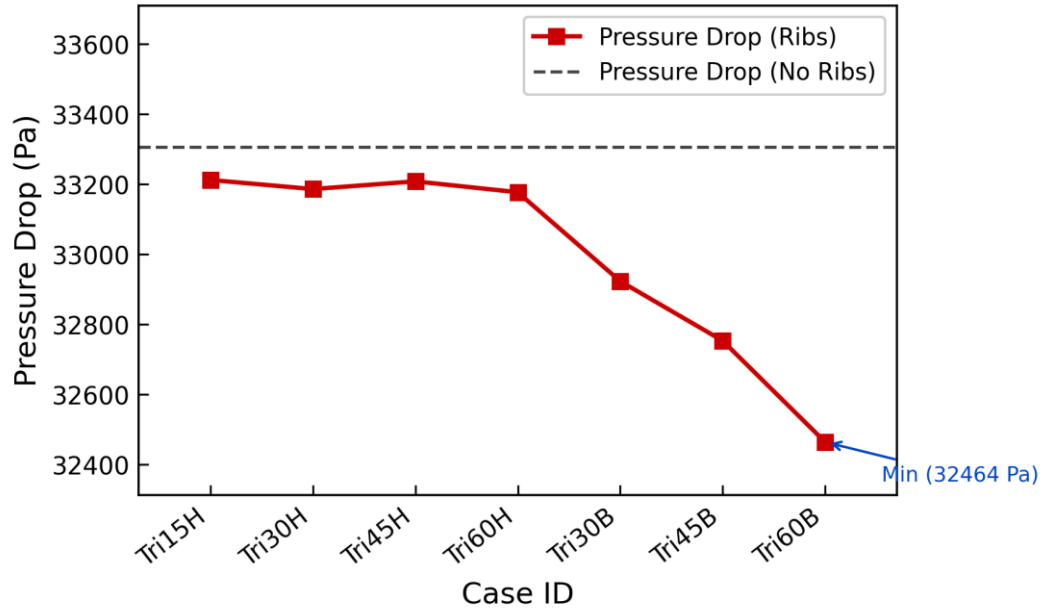


Figure 15: Pressure drop across the oscillator for all ribbed configurations; dashed line shows smooth case baseline

Figure 16 presents the Strouhal number for all configurations, with the Smooth case value of 0.0144 included as a horizontal reference line. The variation in Strouhal number closely follows the oscillation frequency trend, reflecting the direct proportionality between these two parameters. The fixed-height (H) family exhibits Strouhal numbers within a narrow range of 0.0204–0.0207, with both Tri30H-Ribs and Tri60H-Ribs attaining the highest value of 0.0207. In the fixed-base (B) family, Tri30B-Ribs achieves a Strouhal number of 0.0201, whereas Tri45B-Ribs and Tri60B-Ribs record lower values of 0.0167 and 0.0164, respectively. Despite these differences, all ribbed configurations exhibit Strouhal numbers greater than the Smooth baseline, corresponding to improvements of approximately 14-44% over the smooth configuration. These results demonstrate that the incorporation of triangular ribs on the Coanda surfaces consistently enhances the non-dimensional oscillation characteristics of the fluidic oscillator, with the greatest improvement observed in the fixed-height configurations.

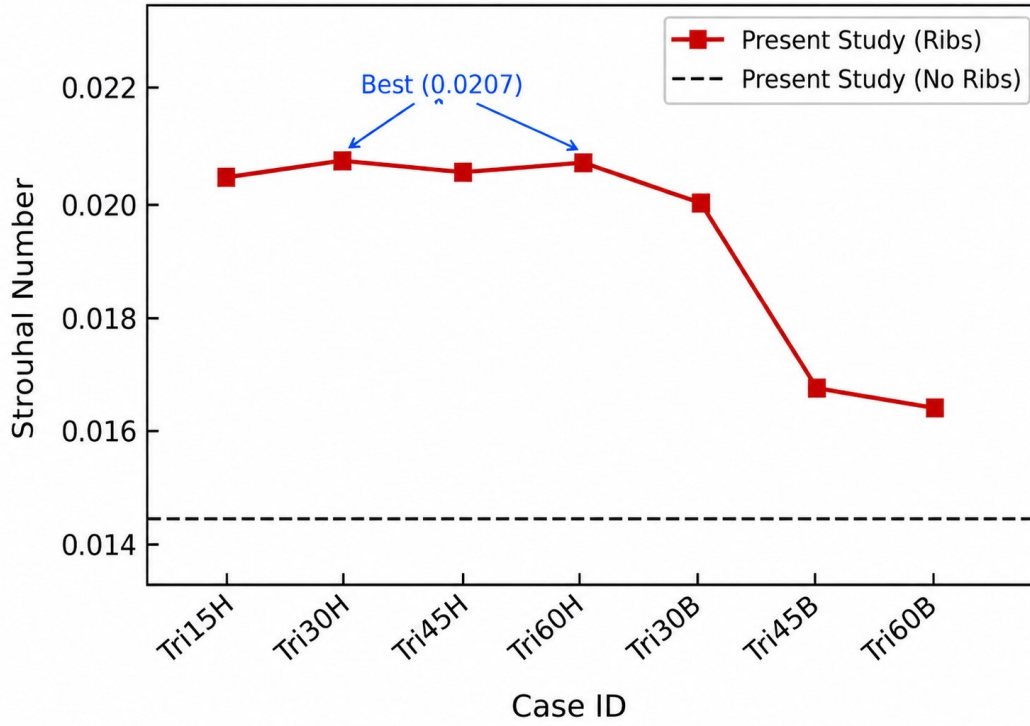


Figure 16: Strouhal number for all ribbed configurations; black line shows smooth case baseline

4.5 Jet Deflection Angle and Jet Deflection-Pressure Ratio

Figure 17 presents the jet deflection angle for all ribbed configurations. Unlike the oscillation frequency, the variation in jet deflection angle does not exhibit a monotonic dependence on the rib geometry. Within the fixed-base family, Tri30B-Ribs preserves a deflection angle of 31.6° , comparable to the Smooth configuration 32.0° , whereas Tri45B-Ribs exhibits a pronounced reduction to 22.0° . A partial recovery is observed for Tri60B-Ribs at 26.7° . Similarly, the fixed-height configurations consistently produce lower deflection angles ranging from 16.7 to 18.9° , despite exhibiting the highest oscillation frequencies. The instantaneous velocity contours in Figure 4 and Figure 5, demonstrate that triangular ribs substantially modify the jet evolution within the mixing chamber. However, they do not reveal a unique flow feature that directly correlates with the measured exit deflection angle. This observation suggests that the exit jet deflection is governed by the cumulative unsteady jet dynamics over an oscillation cycle rather than by a single instantaneous flow characteristic.

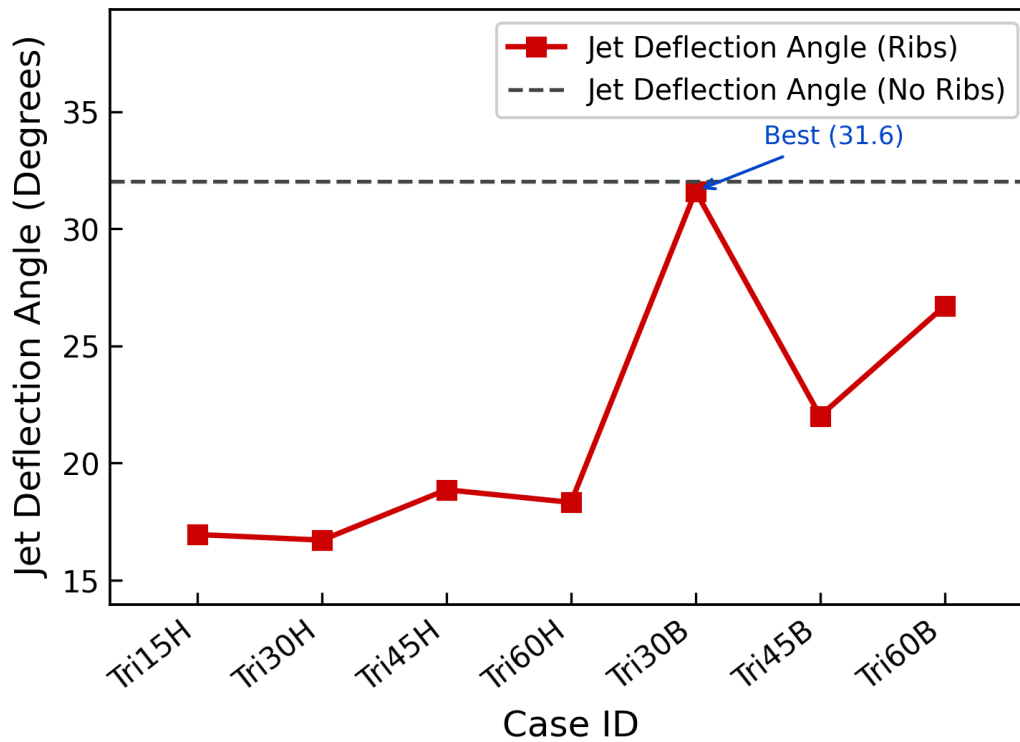


Figure 17: Jet deflection angle for all ribbed configurations; dashed line shows smooth case baseline

Figure 18 presents the JDPR results, which evaluate the jet deflection performance while accounting for the pressure drop penalty by normalizing the jet deflection angle with the cube root of the pressure drop ratio relative to the Smooth configuration. The Smooth case serves as the reference with a JDPR of 1.0. Among the ribbed configurations, Tri30B-Ribs achieves the highest JDPR of 0.991, indicating that its jet deflection performance is nearly equivalent to that of the Smooth configuration despite the geometric modifications. Within the fixed-base family, the JDPR decreases to 0.691 for Tri45B-Ribs before partially recovering to 0.841 for Tri60B-Ribs, reflecting the corresponding variation in jet deflection angle. In contrast, the fixed-height family exhibits consistently lower JDPR values, ranging from 0.522 for Tri30H-Ribs to 0.590 for Tri45H-Ribs, owing to the reduced jet deflection angles observed in these configurations. Overall, the JDPR results indicate that Tri30B-Ribs provides the most balanced compromise between jet deflection capability and pressure drop among the ribbed configurations, while the fixed-height family prioritizes oscillation enhancement at the expense of jet deflection performance.

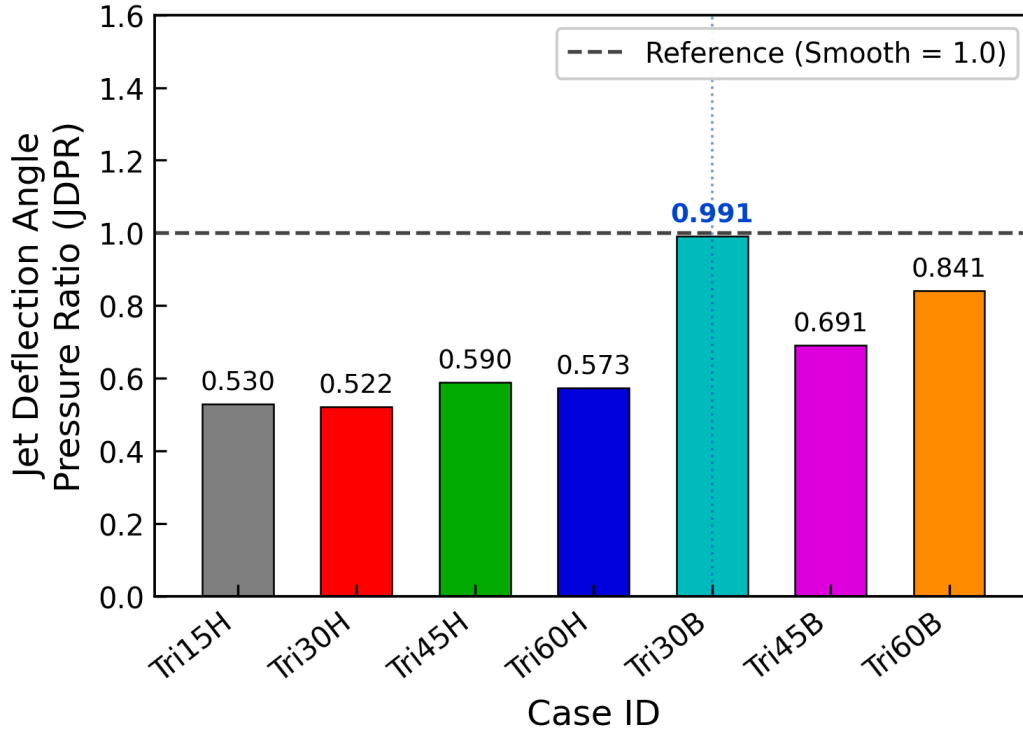


Figure 18: Jet Deflection Angle-Pressure Ratio (JDPR) for all ribbed configurations; dashed line shows smooth case reference

4.6 Combined Performance: Frequency-Deflection-Pressure Ratio

The FDPR simultaneously evaluates oscillation frequency, jet deflection angle, and pressure efficiency relative to the Smooth configuration, which serves as the reference with an FDPR of 1.0. The FDPR values for all ribbed configurations are presented in Figure 19. Among all cases, Tri30B-Ribs achieved the highest FDPR of 1.396, making it the only ribbed configuration to exceed the Smooth baseline. This performance results from the combined effect of a 41% increase in oscillation frequency, a jet deflection angle of 31.6° that closely matches the Smooth configuration, and a reduced pressure drop. In contrast, the remaining configurations exhibit FDPR values below unity, indicating that the improvements in one or more performance metrics are offset by reductions in others. The fixed-height family exhibits FDPR values ranging from 0.768 to 0.861. Although these configurations achieve the highest oscillation frequencies in the present study, their substantially lower jet deflection angles (16.7 - 18.9°) reduce the overall FDPR, demonstrating that the gain in oscillation frequency is insufficient to compensate for the loss in lateral jet coverage within the combined performance metric. The relatively small variation in FDPR among the fixed-height cases is consistent with their similar oscillation frequencies and jet deflection angles. Within this family, Tri45H-Ribs achieved the highest FDPR (0.861), followed by Tri60H-Ribs (0.842), Tri15H-Ribs (0.769), and Tri30H-Ribs (0.768).

Within the fixed-base family, Tri60B-Ribs achieved an FDPR of 0.938, followed by Tri45B-Ribs with 0.809. Although both configurations exhibit lower oscillation frequencies than the fixed-height family, their comparatively larger jet deflection angles improve the overall balance among frequency, jet coverage, and pressure efficiency. The overall FDPR ranking is Tri30B-Ribs (1.396) > Tri60B-Ribs (0.938) > Tri45H-Ribs (0.861) > Tri60H-Ribs (0.842) > Tri45B-Ribs (0.809) > Tri15H-Ribs (0.769) > Tri30H-Ribs (0.768). These results indicate that Tri30B-Ribs provides the most balanced overall flow performance, achieving simultaneous improvements in oscillation frequency, jet deflection, and pressure efficiency relative to the Smooth configuration.

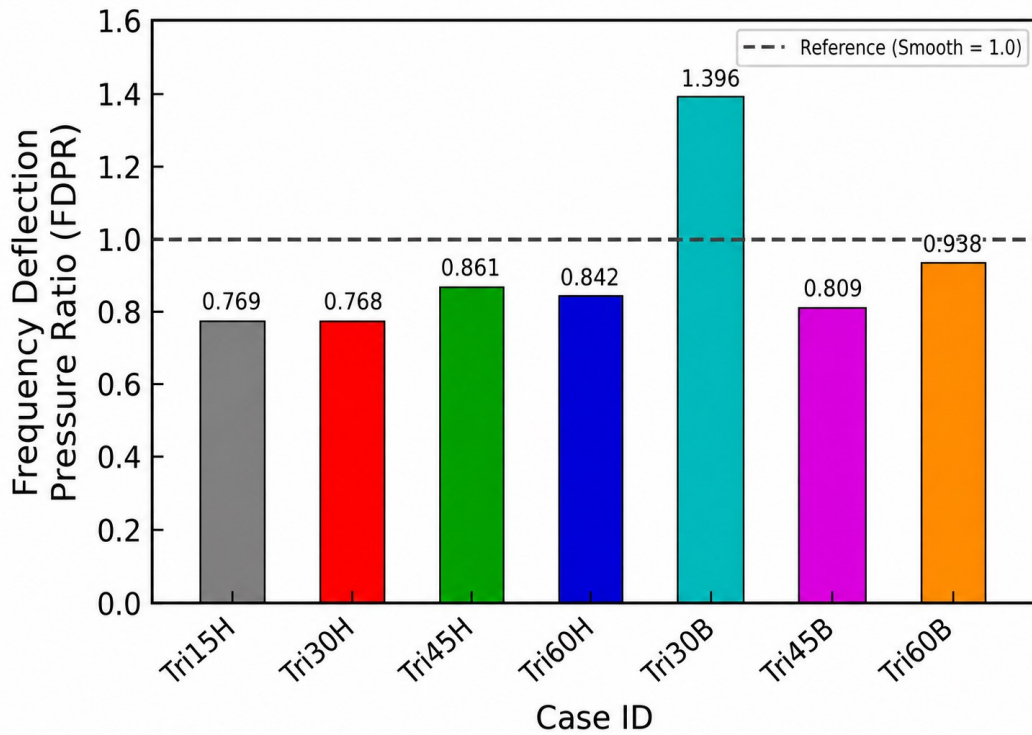


Figure 19: Frequency-Deflection-Pressure Ratio (FDPR) for all ribbed configurations; dashed line shows smooth case reference

4.7 Engineering Significance of Triangular Rib Parameterization

The present investigation provides a systematic assessment of how triangular rib geometry influences the flow performance of a double-feedback fluidic oscillator and establishes practical design guidance for selecting rib configurations based on application-specific performance requirements. The results demonstrate that the two geometric parameterization strategies, fixed-height (H) and fixed-base (B), occupy distinct regions of the design space, reflecting different trade-offs among oscillation frequency, jet deflection, and pressure loss. The fixed-height family is characterized by consistently high oscillation frequencies, achieving improvements of approximately 45–47% over the Smooth configuration across all apex angles investigated. The relatively small variation in oscillation frequency, Strouhal number, recirculation bubble size, and FDPR within this family indicates that the

oscillatory behavior is largely insensitive to apex angle when the rib height is maintained constant. From an engineering perspective, this reduced sensitivity is advantageous because small geometric deviations arising from manufacturing tolerances are unlikely to cause significant variations in oscillator performance. Consequently, the fixed-height parameterization provides a robust design approach for applications in which a high and repeatable oscillation frequency is the primary design objective, such as active flow control, boundary-layer separation control, and enhanced fluid mixing.

In contrast, the fixed-base family exhibits a stronger dependence on apex angle, providing greater flexibility in balancing oscillation frequency, jet deflection, and pressure characteristics. Among all configurations investigated, Tri30B-Ribs achieved the most balanced overall performance, combining a 41% increase in oscillation frequency with a jet deflection angle (31.6°) comparable to that of the Smooth configuration, while simultaneously reducing the pressure drop. These combined improvements resulted in the highest FDPR (1.396) obtained in the present study, indicating the most favorable compromise among the three performance objectives considered. Such a performance balance is particularly attractive for applications in which both oscillation frequency and sweeping jet coverage contribute directly to system effectiveness, including impingement cooling, turbine blade thermal management, and sweeping-jet flow control. The higher apex-angle fixed-base configurations (Tri45B and Tri60B) exhibited lower pressure losses but at the expense of reduced oscillation frequency and jet deflection angle, illustrating the inherent trade-offs associated with modifying the rib geometry. These findings demonstrate that no single geometric parameter can independently optimize all performance characteristics; instead, the overall flow-characteristic response is governed by the coupled influence of rib height, apex angle, and their effects on unsteady jet dynamics. Overall, the proposed two-family design framework provides a physically interpretable methodology for tailoring fluidic oscillator performance to different engineering objectives. Beyond identifying high-performing configurations, the framework establishes a systematic basis for future investigations involving three-dimensional oscillator geometries, broader Reynolds number ranges, heat transfer enhancement, and multi-objective optimization, where oscillation frequency, jet coverage, pressure loss, and thermal performance can be considered simultaneously.

5 Conclusions

The present study numerically investigated the influence of triangular rib parameterization on the flow performance of a double-feedback fluidic oscillator by introducing four triangular ribs on each Coanda surface. Two geometric parameterization strategies, namely fixed-height (FH) and fixed-base (FB), were systematically examined over a range of half apex angles using two-dimensional unsteady compressible CFD with the SST $k-\omega$ turbulence model. The numerical results demonstrate that

triangular rib parameterization significantly enhanced the flow performance of the fluidic oscillator. All ribbed configurations exhibited higher oscillation frequencies and Strouhal numbers than the Smooth configuration, accompanied by a substantial reduction in the primary recirculation bubble within the mixing chamber.

The fixed-base (FB) family provided the greatest flexibility for balancing oscillation frequency, jet deflection angle, and pressure loss. Among all configurations, Tri30B-Ribs achieved the most favorable overall flow performance by combining an oscillation frequency of 599 Hz (41.0% higher than the Smooth configuration) with a jet deflection angle of 31.6° , comparable to the Smooth case (32.0°). Consequently, Tri30B-Ribs achieved the highest JDPR (0.991) and FDPR (1.396), demonstrating the best overall compromise among oscillation frequency, jet coverage, and pressure efficiency. In contrast, the fixed-height (FH) family consistently produced the highest oscillation frequencies, ranging from 617 to 625 Hz, corresponding to improvements of 45.2–47.1% over the Smooth configuration. The narrow variation in oscillation frequency, Strouhal number (0.0204–0.0207), recirculation bubble area (113.95–117.14 mm²), and FDPR indicates that rib height is the dominant geometric parameter governing the oscillation characteristics, whereas variations in apex angle produce only minor changes when the rib height is maintained constant.

The primary recirculation bubble exhibited a clear inverse relationship with oscillation frequency. The Smooth configuration produced the largest bubble area (231.57 mm²), whereas the ribbed configurations reduced the bubble area by approximately 37–51%, thereby promoting more rapid jet switching. In contrast, the exit jet deflection angle did not exhibit a monotonic dependence on rib geometry, indicating that it is governed by the cumulative unsteady jet evolution rather than by a single geometric parameter. The proposed FPR, JDPR, and FDPR metrics provided complementary measures for evaluating the trade-offs among oscillation frequency, jet deflection, and pressure loss. While the FPR highlighted the frequency gain relative to the pressure penalty and the JDPR quantified the jet deflection capability relative to pressure loss, the FDPR integrated all three performance characteristics into a single evaluation metric. These metrics collectively identified Tri30B-Ribs as the most balanced design, whereas the fixed-height family remained the preferred choice for applications where maximizing oscillation frequency is the primary objective.

Overall, the proposed triangular rib parameterization establishes a systematic passive design framework for tailoring the flow performance of double-feedback fluidic oscillators through Coanda surface modification. The findings provide practical engineering guidance for selecting rib geometries according to application-specific performance requirements and establish a foundation for future investigations involving three-dimensional oscillator geometries, heat transfer enhancement, broader Reynolds number ranges, and multi-objective optimization.

Data Availability Statement: Data available on request from the authors.

Generative AI Use Statement: ChatGPT (OpenAI) was used solely for language editing, grammar, and readability improvement. It was not used to generate or develop any original scholarly, intellectual, substantive, or scientific content of this work. All research, analysis, interpretations, results, and conclusions are the authors' own.

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